IDAT7215
Computer Programming for Product Development and Applications

Lecture 3-1: Introduction to Artificial Intelligence

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Outline

- What is AI?
- History of AI
- Application of AI
- Search Algorithms for AI

What is intelligence

- The ability to learn or understand from experience.
- The ability to acquire and retain knowledge.
- The ability to respond quickly and successfully to a new situation.
- The ability to use reason to solve problems.

What is human intelligence

- It is a composition of abilities like



Learning



Reasoning



Perceiving



Understanding of Language



Feeling

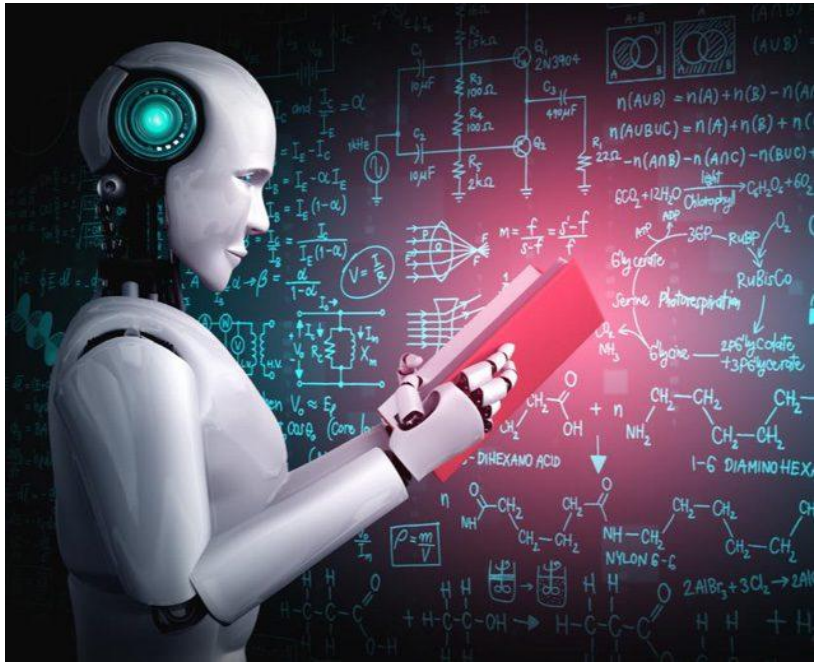
- If intelligence is learning, understanding, retaining, responding, and using reason than what is Artificial Intelligence (AI) ?

Quick Answer from Academia

- Modeling human cognition using computers.
- Study of making computer do things which at the moment humans are better at..
- Making computers do things which require intelligence.

What is Artificial Intelligence (AI)

- **Artificial Intelligence (AI):** The simulation of human intelligence by machines.
  - The ability to solve problems.
  - The ability to act rationally.
  - The ability to act like humans.



- Computers with the ability to mimic or duplicate the functions of human brain.

- The central principles of AI include:
  - Reasoning, knowledge, planning, learning, and communication.
  - Perception and the ability to move and manipulate objects.
  - It is the science and engineering of making intelligent machines. Especially intelligent computer.

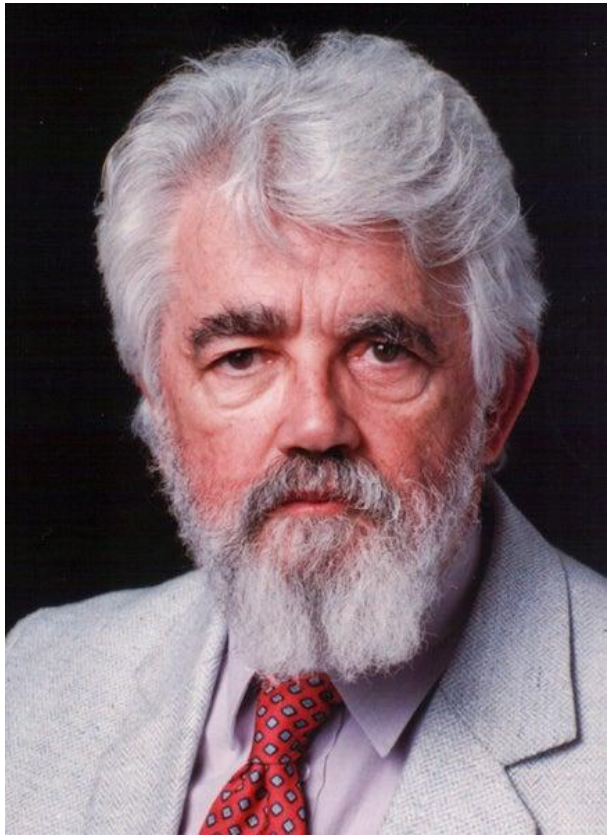


What's Involved in Intelligence?

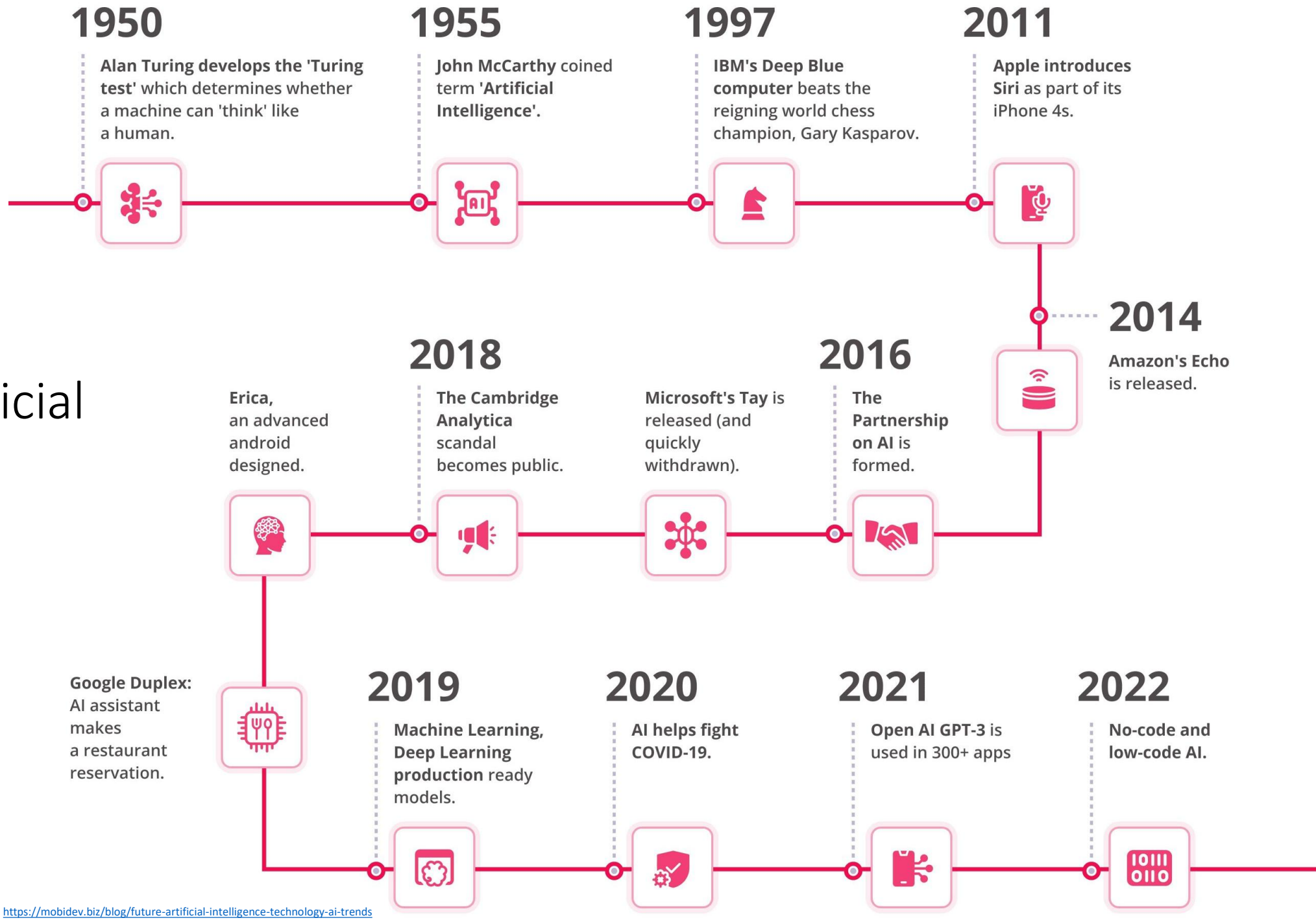
- Ability to interact with the real world
  - To perceive, understand, and act
    - ✓ E.g., speech recognition and understanding
- Searching the best solution
- Reasoning and Planning
  - Modelling the external world-delivery robot
  - Solving new problems, planning, and making decisions.
  - Ability to deal with unexpected problems, uncertainties.
- Learning and Adaption
  - We are continuously learning and adapting.
  - Our internal models are always being updated.
    - ✓ E.g., a baby learning to categorize and recognize animals.

John McCarthy

- (September 4, 1927 – October 24, 2011) was an American Computer Scientist and Cognitive Scientist.
- McCarthy was one of the founders of the discipline of Artificial Intelligence.
- He coined the term “ Artificial Intelligence (AI) “



History of Artificial Intelligence



https://mobidev.biz/blog/future-artificial-intelligence-technology-ai-trends

A Brief History of AI



https://www.youtube.com/watch?v=yaL5ZMvRRqE&t=338s&ab_channel=PlattformLernendeSysteme

Views of AI fall into four categories in Two dimensions:

- **Two dimensions**
  - **Thinking**/Reasoning vs **Behavior**/Action
  - Success according to **human** standards vs Success according to ideal concept of intelligence (**rationality**):
- **Four Categories:**
  - **Systems that think like humans** (focus on reasoning and human framework).
  - **Systems that think rationally** (focus on reasoning and general concept of intelligence).
  - **Systems that act like humans** (focus on behavior and human framework).
  - **System that act rationally** (focus on behavior and general concept of intelligence).

Definitions of artificial intelligence (AI)?

	HUMAN	RATIONAL
THINKING	**1. Thinking humanly** The exciting new effort to make computers think.. Machine with minds ..(Hugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning..." (Bellman, 1978)	**2. Thinking rationally** "The study of computations that make it possible to perceive, reason, and act" (Wintson, 1992) "The study of mental faculties through the use of computational models", (Charniak and McDermott, 1985)
BEHAVIOUR	**3. Acting humanly** "The art of creating machines that perform functions that require intelligence when performed by people" (Kurzwell, 1990) "The study of how to make computers do things at which, at the moment, people are better" (Rich and Knight, 1991)	**4. Acting rationally** "A field of study that seeks to explain and emulate intelligent behavior in terms of computational process" (Schalkoff, 1990) "The branch of computer science that is concerned with the automation of intelligent behaviour" (Luger and Stubblefield, 1993)

The Foundation of AI

- **AI is a young field**
  - Inherit many ideas, viewpoints, and techniques from other disciplines:
    - ✓ Philosophy
      - ➢ Considering the ideas that mind is in some ways like a machine
      - ➢ Logic, foundations of learning, language
    - ✓ Mathematics
      - ➢ Formal representation and proof
      - ➢ Algorithms
    - ✓ Psychology
      - ➢ Strengthen the idea that humans and other animals can be considered information processing machines.
    - ✓ Linguistics
    - ✓ Computer engineering

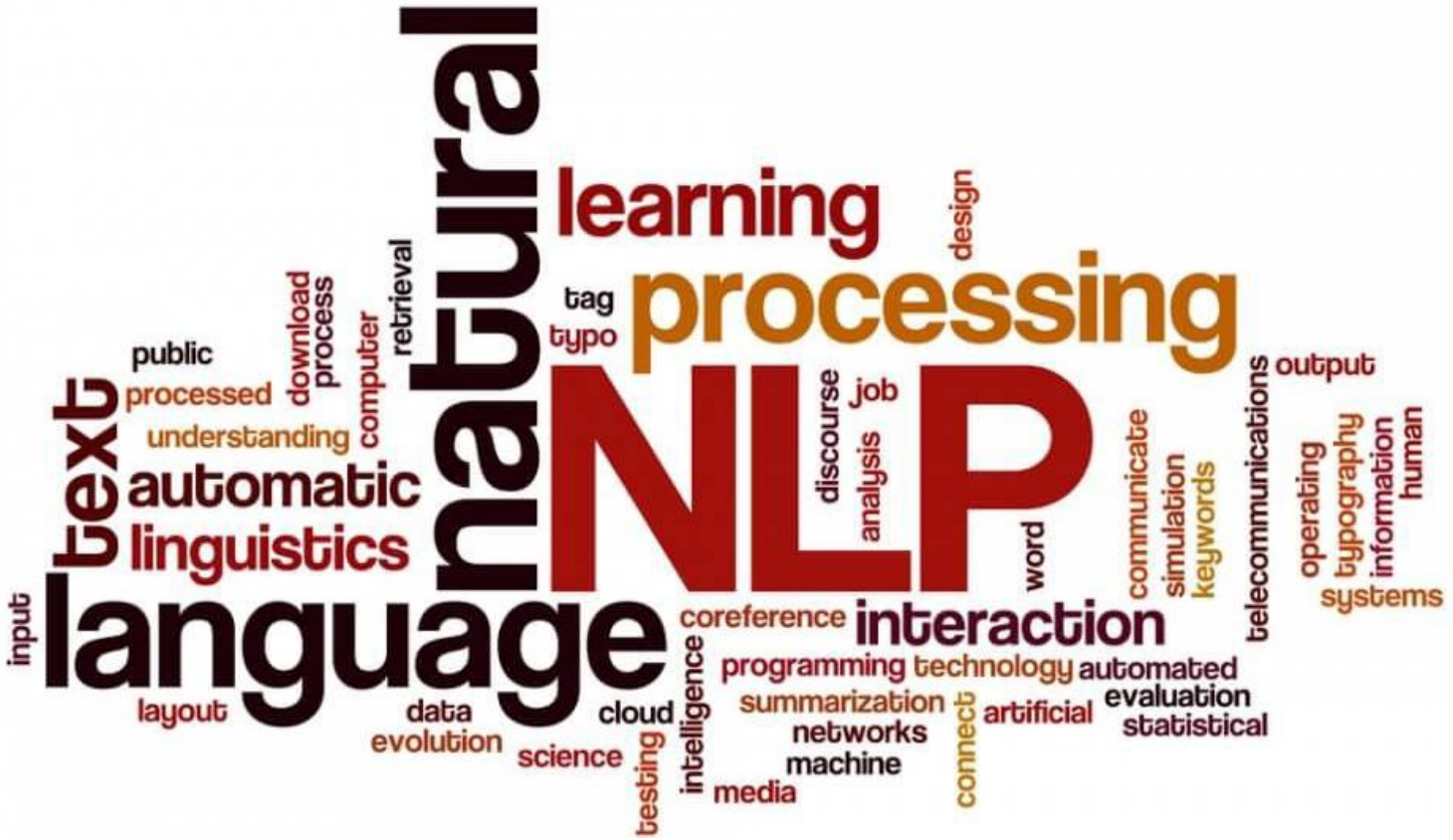
- **Symbolic AI** is concerned with describing and manipulating our knowledge of the world as explicit symbols, where these symbols have clear relationships to entities in the real world.
- **Sub-symbolic AI** (e.g. neural-nets) is more concerned with obtaining the correct response to an input stimulus without 'looking inside the box to see if parts of the mechanism can be associated with discrete real-world objects.

Application of AI

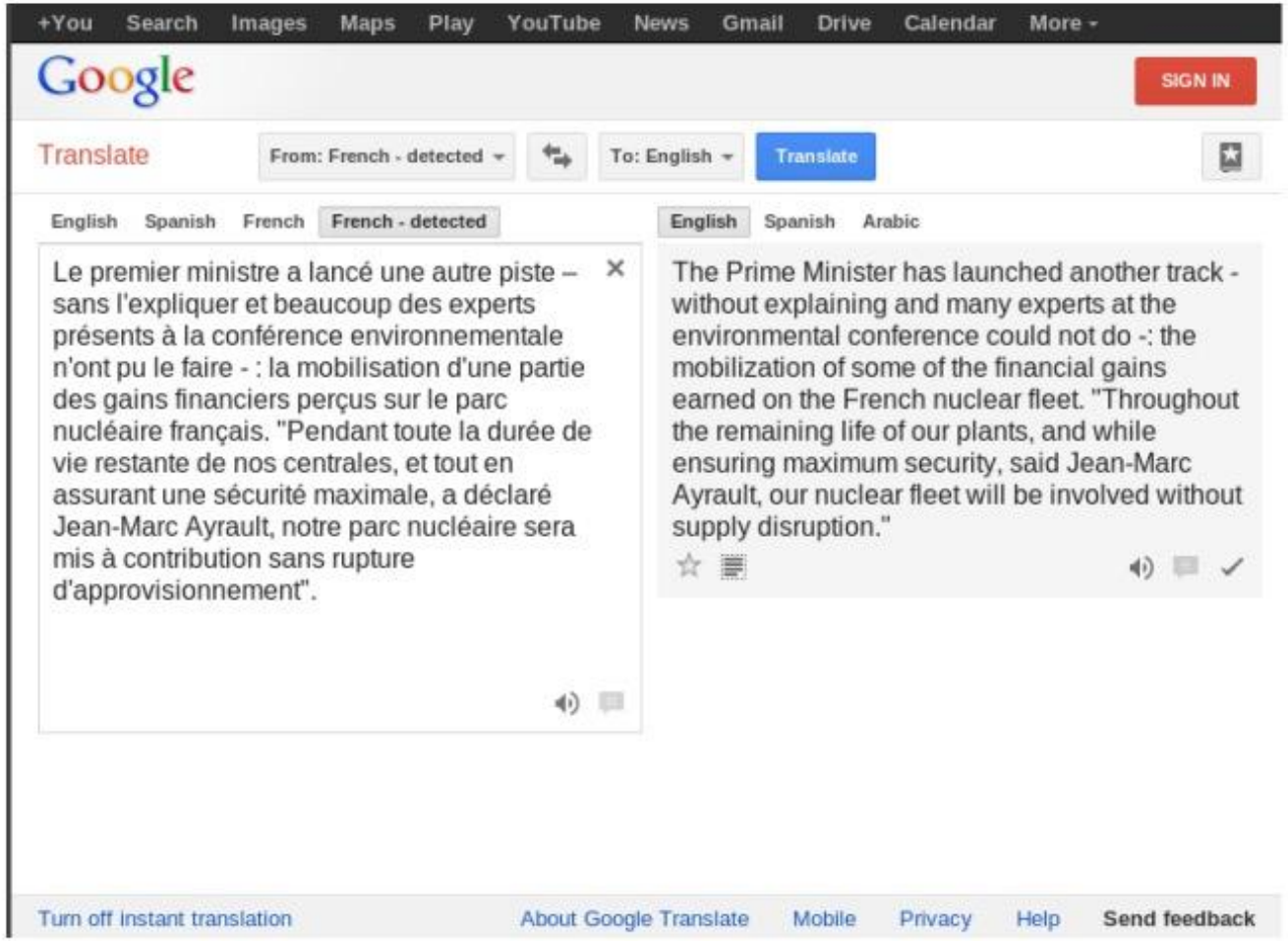
- Natural Language Understanding
- Expert System
- Planning and Robotics
- Machine Learning
- Game Playing

Natural Language Processing

- To design and build software that will analyze understand and generate languages that humans use naturally.



Machine Translation



- Machine translation research started in the 1960s (the US government was quite keen on translating Russian into English). Over the subsequent decades, it went through quite a few rough turns.
- In the 1990s and 2000s, statistical machine translation, aided by large amounts of example human translations, helped vastly improve translation quality.
- As of 2015, Google Translate supports 90 languages and serves over 200 million people daily. The translations are nowhere near perfect, but they are very useful.
- In 2016, Google released a neural machine translation system which led to significant improvements in accuracy due to advances in deep learning, and as of July 2022, 133 languages are supported.

Reading comprehension

- Natural language understanding generally is still widely regarded as an unsolved problem. One of the specific examples is the task of reading comprehension: given a passage, the goal is to answer a question about the passage (think standardized tests).

In meteorology, precipitations is any product of the condensation of atmospheric water vapor that falls under gravity. The main forms of precipitation include drizzle, rain, sleet, snow, graupel and hail… Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals within a cloud. Short, intense periods of rain in scattered locations are called “showers”.

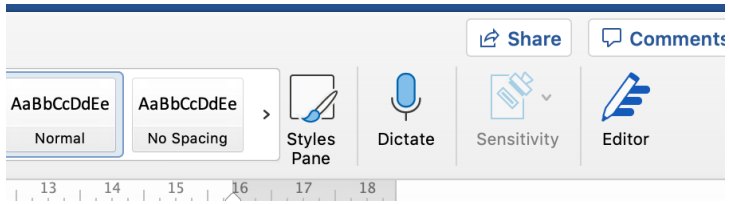
What causes precipitation to fall?
gravity

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?
graupel

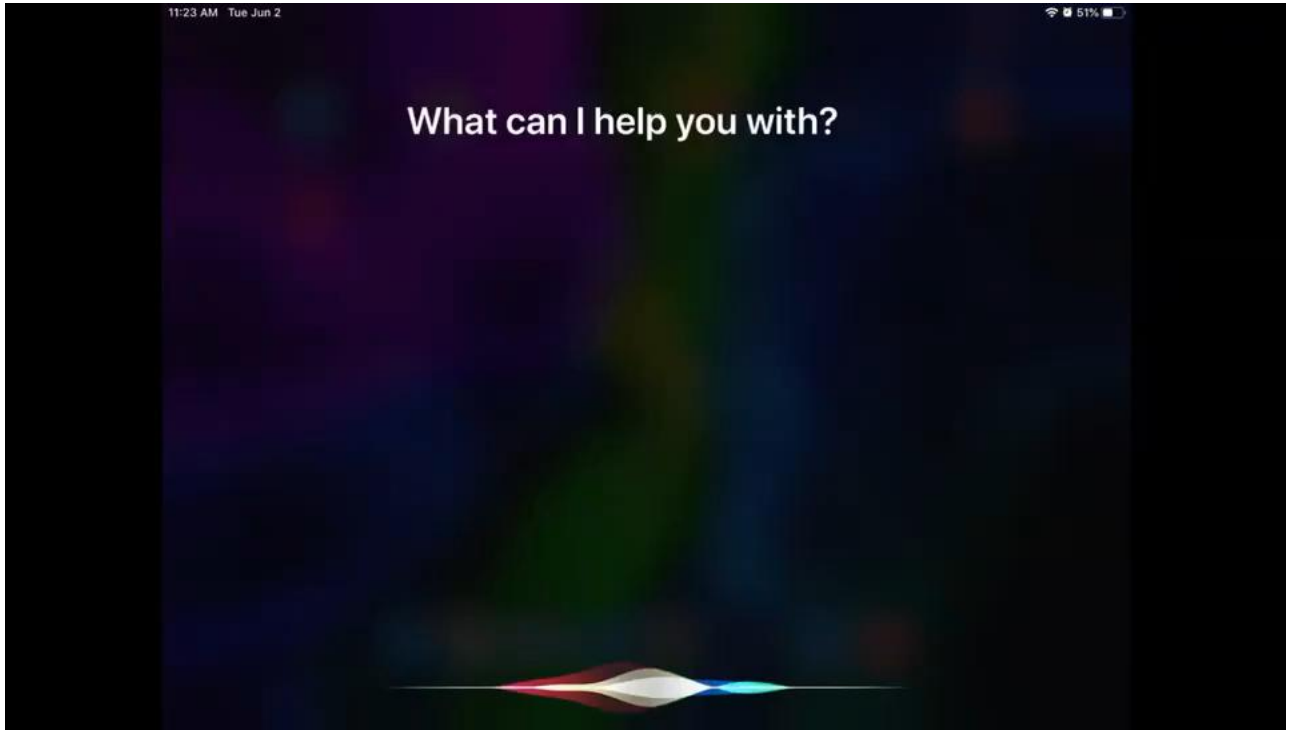
Where do water droplets collide with ice crystals to form precipitation?
within a cloud

Speech recognition

- Speech recognition is the problem of transcribing audio into words.
- It has a long history dating back to the 1970s.
- But it wasn't until around 2009 that speech recognition started to work well due to the adoption of deep neural networks.



However, speech recognition is only one part of the story; the other is understanding the text, which is a much harder problem.



https://www.youtube.com/watch?v=PfHo6Ad0RYg

Autonomous driving

- Perhaps the first significant event was the 2005 DARPA Grand Challenge, in which the goal was to have a driverless car go through a 132-mile off-road course. Stanford finished in first place. The car was equipped with various sensors (laser, vision, radar), whose readings needed to be synthesized to localize the car and then generate control signals for the steering, throttle, and brake.

- In 2007, DARPA created an even harder Urban Challenge, which was won by CMU.

- In 2009, Google started a self-driving car program, and since then, their self-driving cars have driven over 1 million miles on freeways and streets.

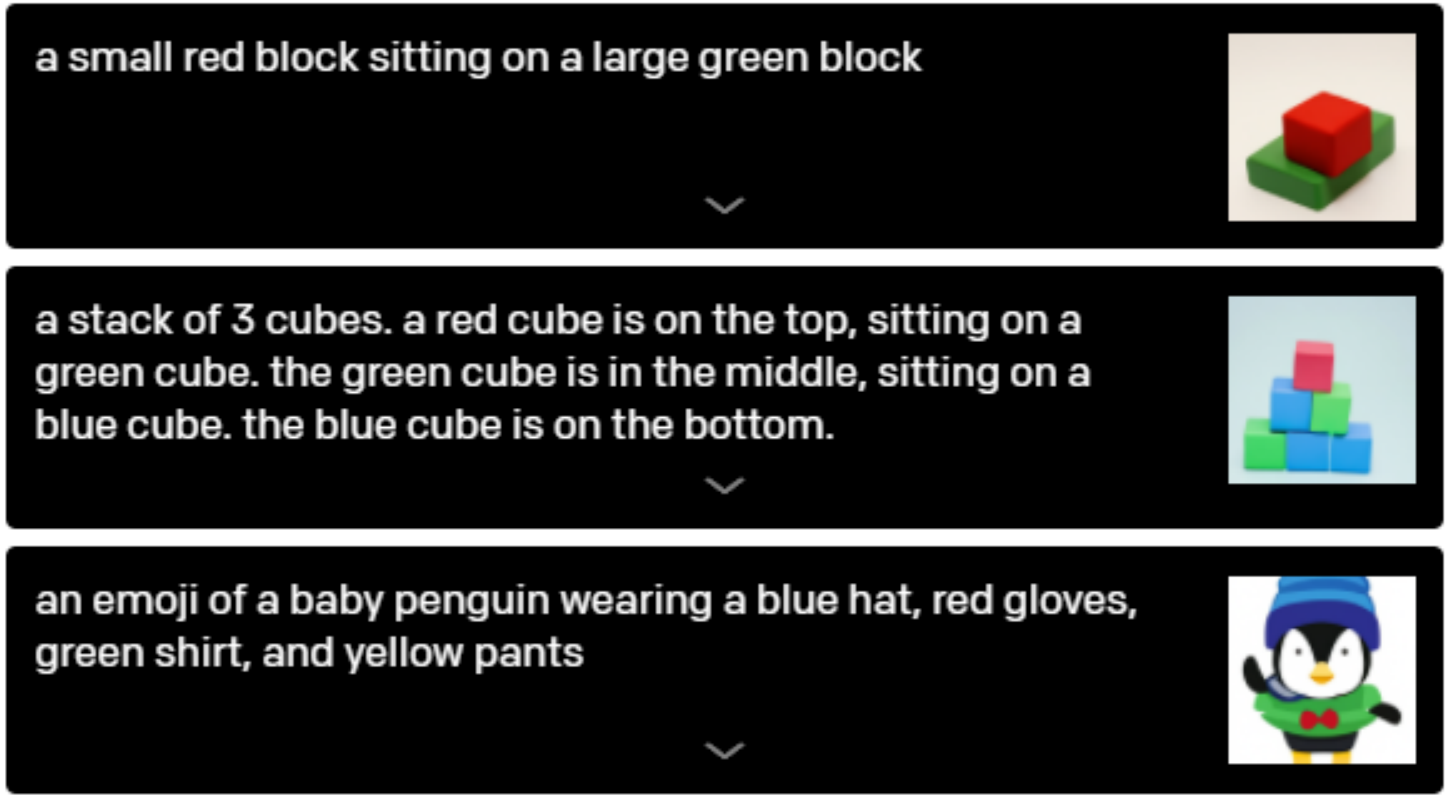
- In January 2015, Uber hired about 50 people from CMU's robotics department to build self-driving cars.



https://www.youtube.com/watch?v=tlThdr3O5Qo

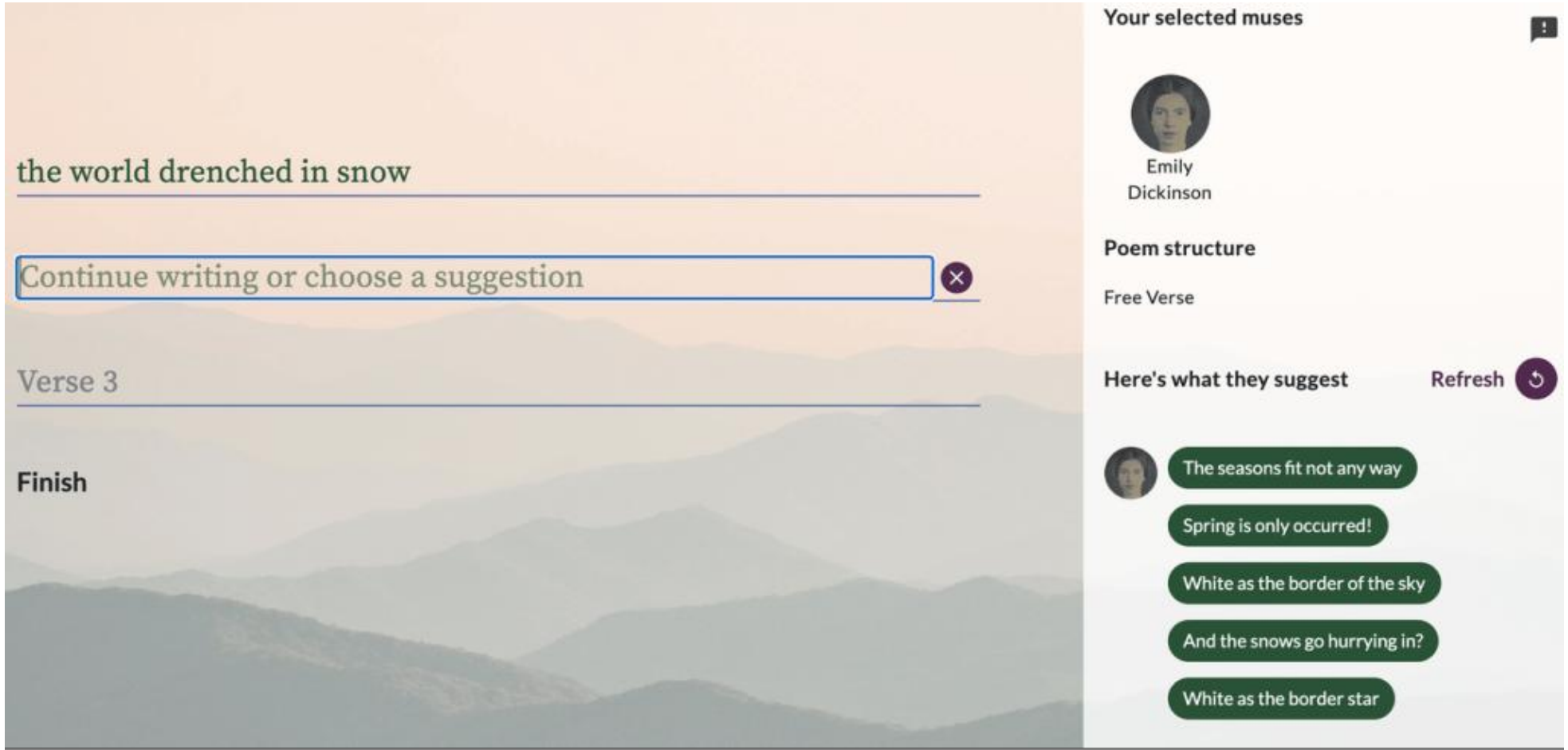
Image generation

- One particular hot topic in computer vision right now is generating photorealistic images from text.



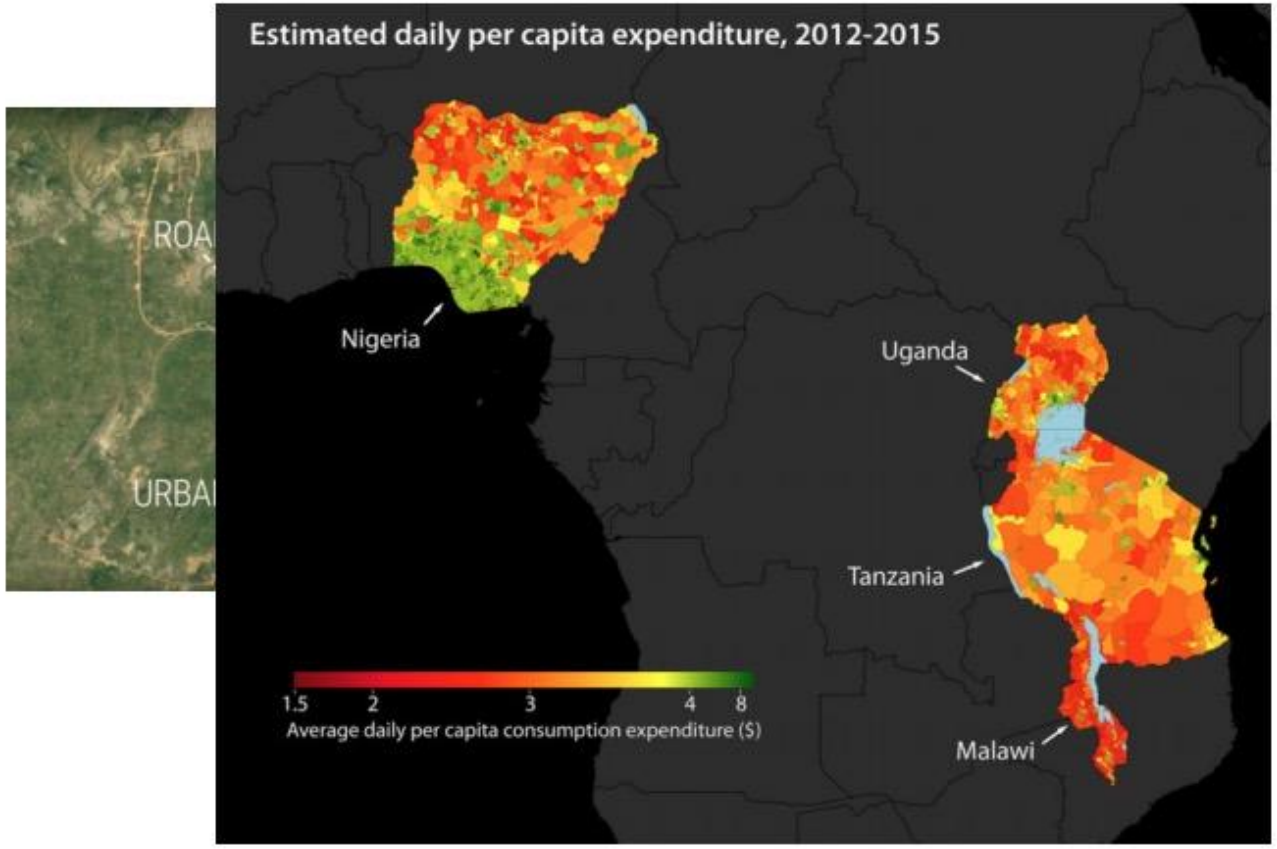
https://openai.com/blog/dall-e/

Poetry writing



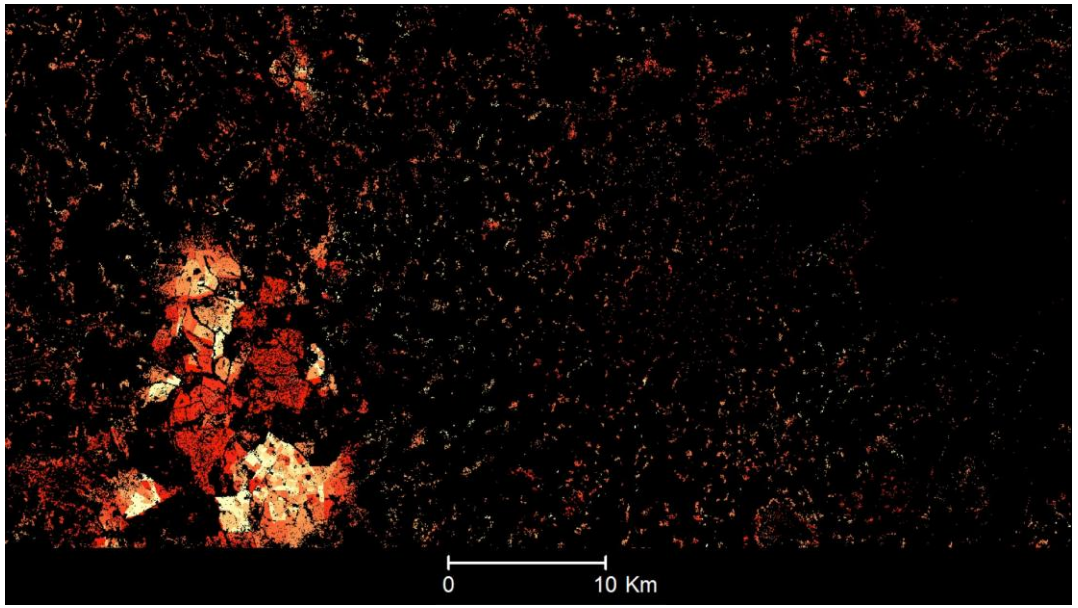
https://sites.research.google/versebyverse/

Tackle social problems



Predicting poverty





A high-resolution population density map layer for Pretoria, South Africa.

Artificial Intelligence for IT Operations

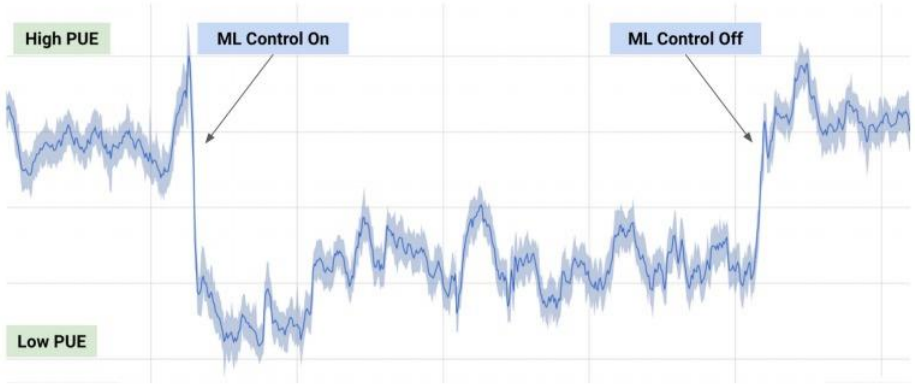
- Saving energy by cooling data centers



Applied

DeepMind AI Reduces Google Data Centre Cooling Bill by 40%

July 20, 2016



https://www.youtube.com/watch?v=Pftge_ewxYQ&ab_channel=RajamanickamAntonimuthu

Humans versus machines

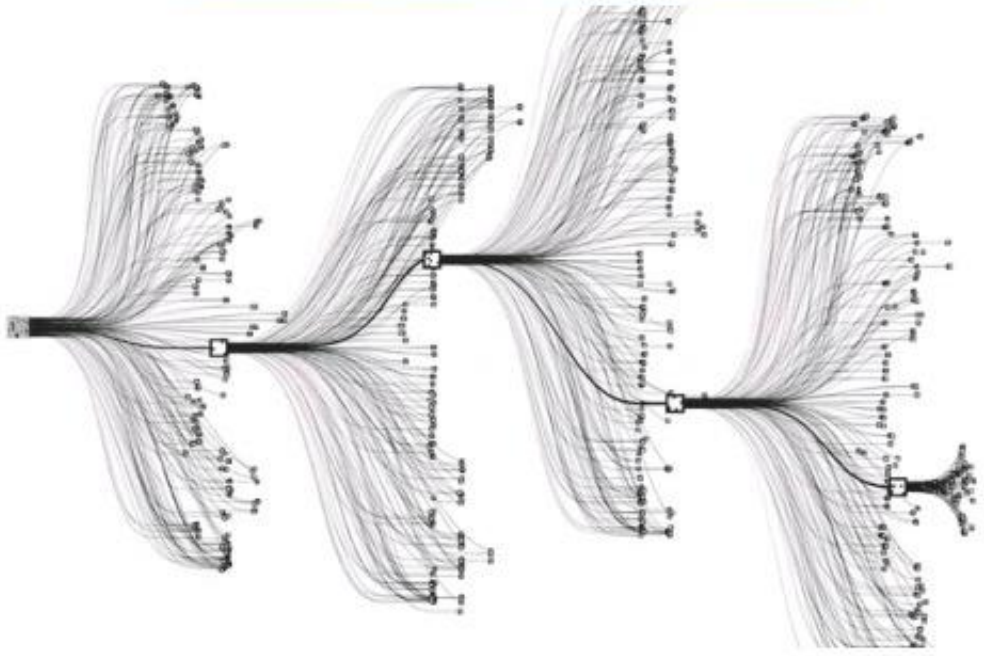


1997: Deep Blue (chess)



2011: IBM Watson (Jeopardy!)





- In 1997, Deep Blue defeated the world chess champion.
- In 2011, IBM Watson defeated two of the biggest winners at the quiz show Jeopardy
- In 2016, Google DeepMind created a program called AlphaGo, which used deep neural networks and reinforcement learning, to defeat a world champion, surprising not only the master Go player but many AI researchers as well.

Some failures...

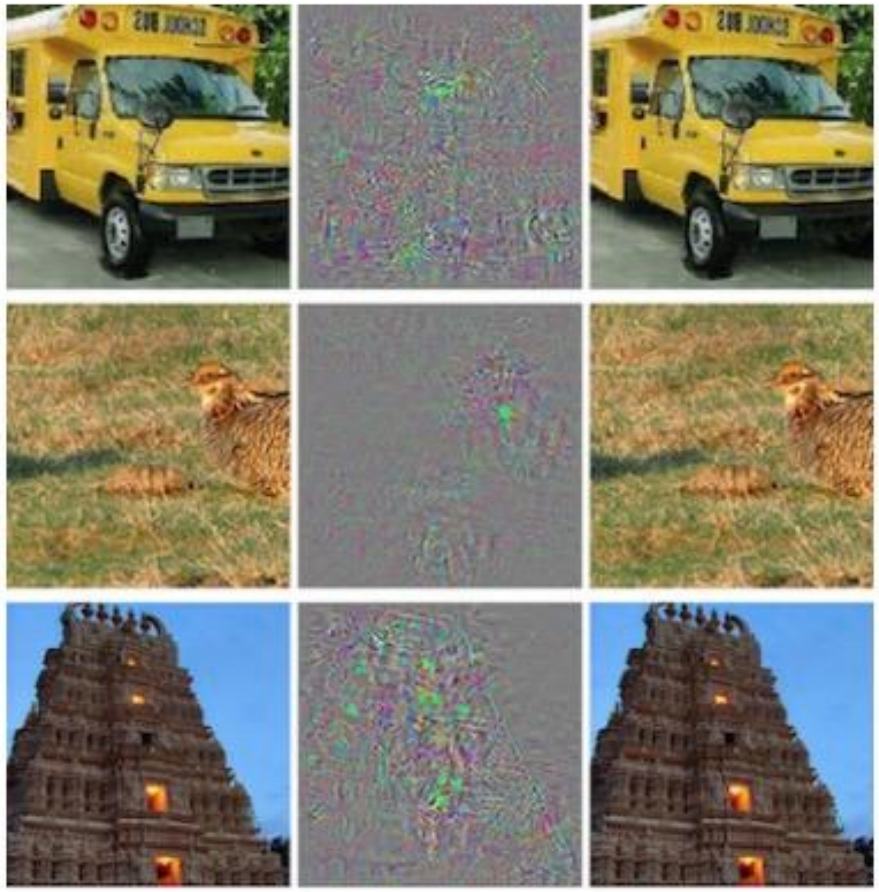
Open-domain dialogue

- A: How old are you?
- B: I’m 16. Why are you asking?
- A: I thought you were 12.
- B: What made you think so?
- A: I don’t know what you are talking about.
- B: You don’t know what you are saying.

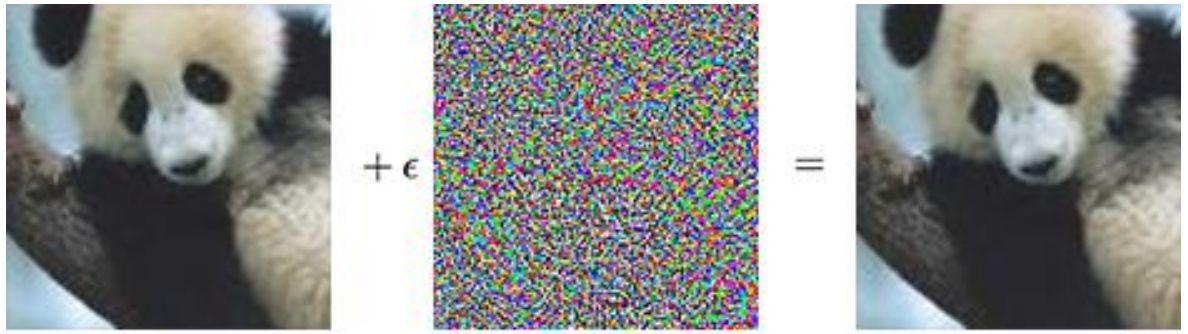


Adversarial examples

AlexNet predicts correctly on the left



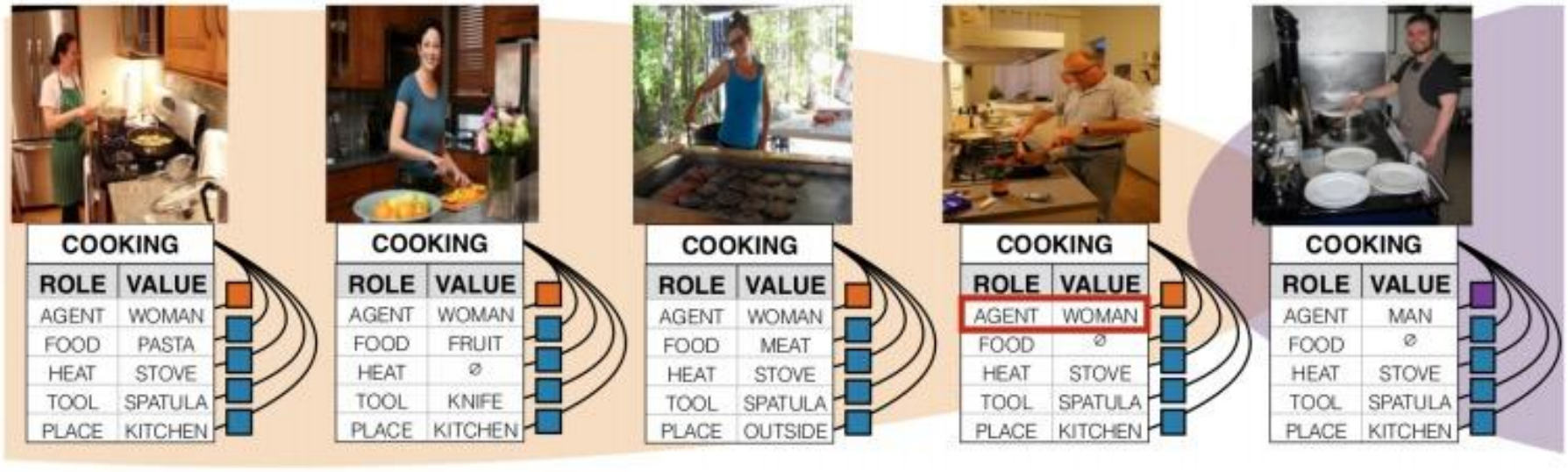
AlexNet predicts **ostrich** on the right

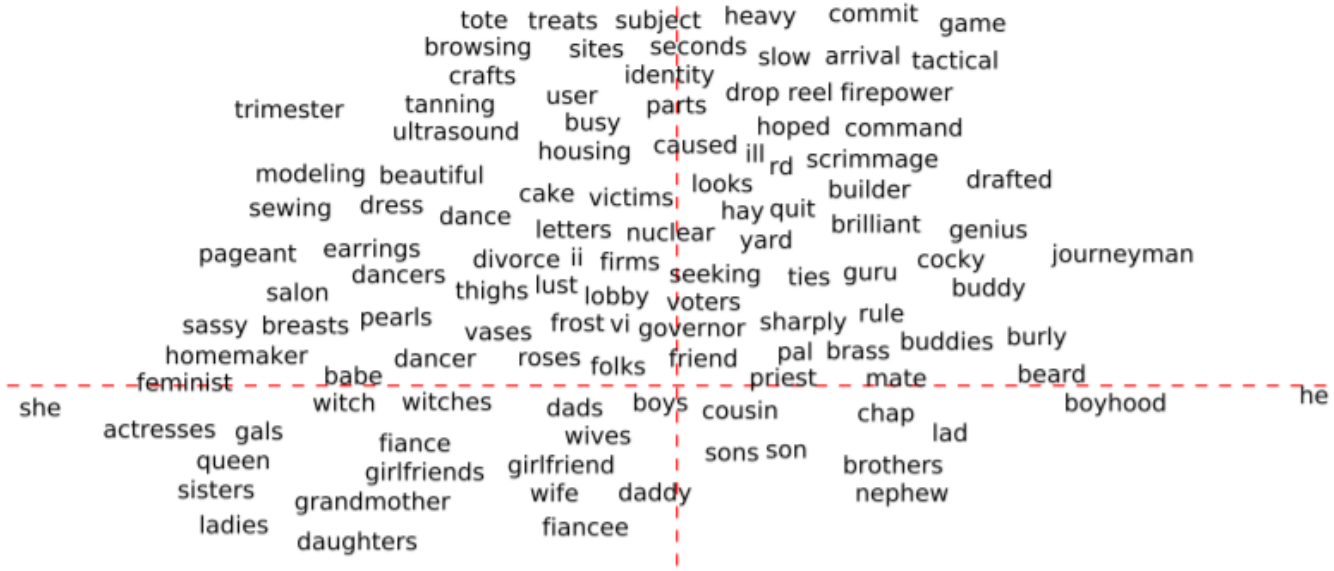


"panda"
57.7% confidence

"gibbon"
99.3% confidence

Bias


COOKING
ROLE VALUE
AGENT WOMAN
FOOD PASTA
HEAT STOVE
TOOL SPATULA
PLACE KITCHEN
COOKING
ROLE VALUE
AGENT WOMAN
FOOD FRUIT
HEAT ∅
TOOL KNIFE
PLACE KITCHEN
COOKING
ROLE VALUE
AGENT WOMAN
FOOD MEAT
HEAT STOVE
TOOL SPATULA
PLACE OUTSIDE
COOKING
ROLE VALUE
AGENT WOMAN
FOOD ∅
HEAT STOVE
TOOL SPATULA
PLACE KITCHEN
COOKING
ROLE VALUE
AGENT MAN
FOOD ∅
HEAT STOVE
TOOL SPATULA
PLACE KITCHEN


she
feminist
homemaker
sassy
salon
pageant
modeling
sewing
trimester
ultrasound
crafts
browsing
tote
he
boyhood
beard
mate
priest
buddies
burly
brass
pal
rule
sharply
governor
frost
vases
pearls
breasts
babe
dancer
roses
folks
friend
witch
witches
dads
boys
cousin
chap
lad
brothers
nephew
son
sons
wives
girlfriend
wife
daddy
fiancee
fiance
girlfriends
grandmother
daughters
queen
sisters
ladies
actresses
gals
dancers
earrings
dress
dance
beautiful
thighs
lust
lobby
voters
divorce
ii
firms
seeking
ties
guru
cocky
journeyman
buddy
letters
nuclear
yard
brilliant
genius
cake
victims
looks
builder
drafted
hay
quit
housing
caused
ill
rd
scrimmage
busy
hoped
command
user
parts
drop
reel
firepower
tanning
identity
seconds
slow
arrival
tactical
sites
subject
heavy
commit
game
treats

Privacy

In 2014, Facebook Research published a paper describing their DeepFace face identification system. DeepFace is a 120 million parameter deep neural network and obtains 97.35% on the standard Labeled Faces in the Wild (LFW) dataset, which is comparable to human performance.

!!! it would enable some entities to take arbitrary images and videos of crowds and identify every single person identity, which would effectively eliminate the ability to stay anonymous.



https://www.youtube.com/watch?v=umxGfNQmDbs

Privacy



Royal Free breached UK data law in 1.6m patient deal with Google's DeepMind

Information Commissioner's Office rules record transfer from London hospital to AI company failed to comply with Data Protection Act



▲ 'We underestimated the complexity of the NHS and of the rules around patient data' – DeepMind. Photograph: Alamy Stock Photo

London's Royal Free hospital failed to comply with the Data Protection Act when it handed over personal data of 1.6 million patients to DeepMind, a Google subsidiary, according to the Information Commissioner's Office.

In the spotlight…

Companies

Google "An important shift from a mobile first world to an AI first world" [CEO Sundar Pichai @ Google I/O 2017]

Microsoft Created AI and Research group as 4th engineering division, now 8K people [2016]

facebook Created Facebook AI Research, Mark Zuckerberg very optimistic and invested

Others: IBM, Amazon, Apple, Uber, Salesforce, Baidu, Tencent, etc.

Governments

 "AI holds the potential to be a major driver of economic growth and social progress" [White House report, 2016]

 Released domestic strategic plan to become world leader in AI by 2030 [2017]

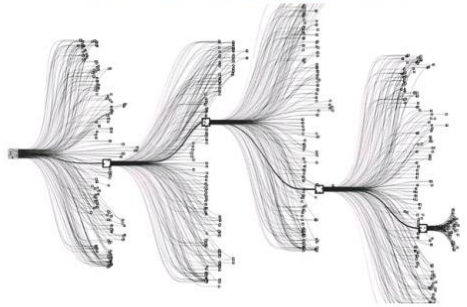
 "Whoever becomes the leader in this sphere [AI] will become the ruler of the world" [Putin, 2017]

Characteristics of AI tasks

- High societal impact (affect billions of people)
- Diverse (language, games, robotics)
- Complex (really hard)

Two sources of complexity...





Computational complexity: exponential explosion

e.g., in the game of Go, there are up to 361 legal moves per turn, and let us say that the average game is about 200 turns. Then, as a crude calculation, there might be 361,200 game trajectories that a player would have to consider to play optimally.

Computation (time/memory)



这是什么意思?



Even infinite computation isn't enough...need to somehow *know* stuff.

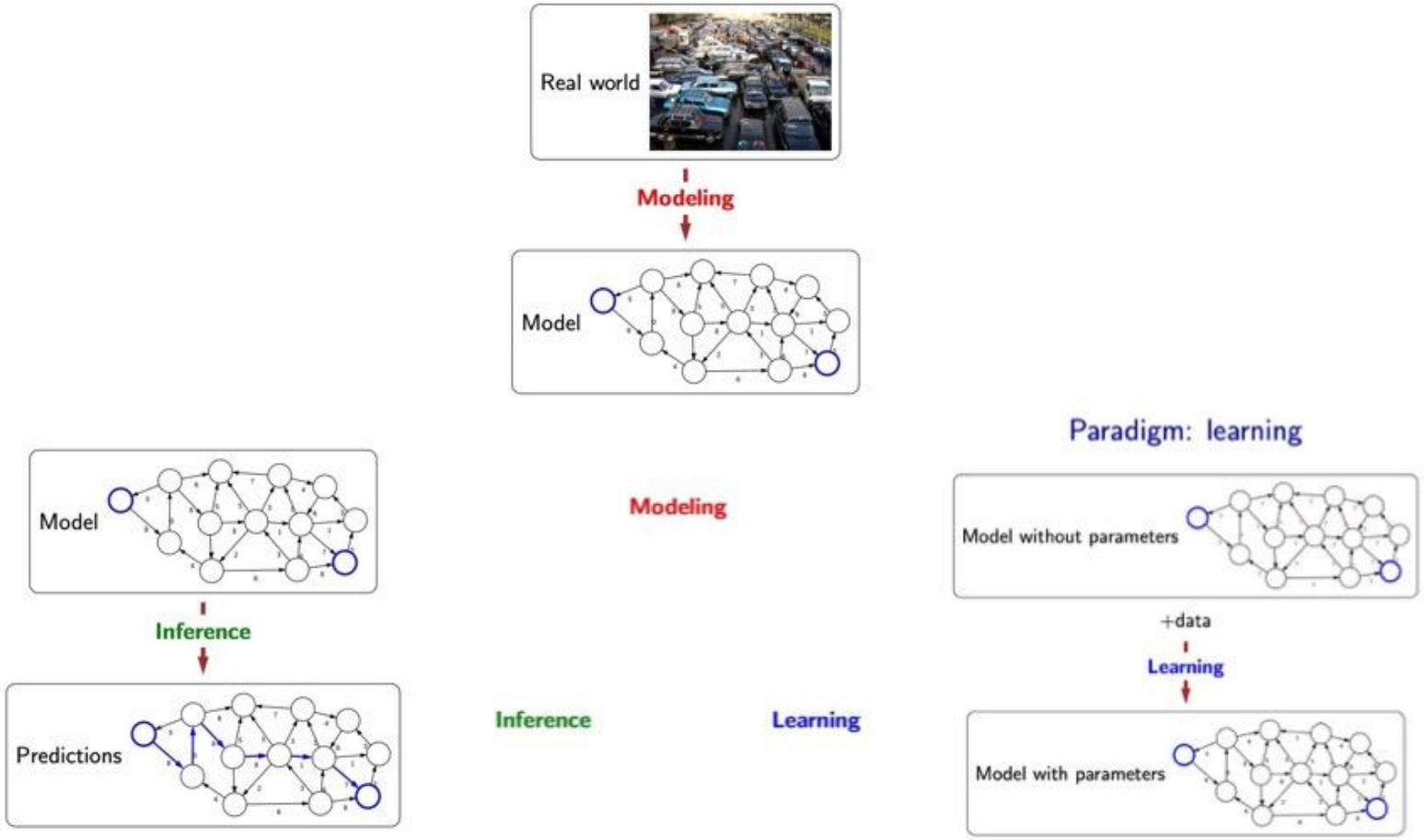
Information complexity: need to acquire knowledge

e.g., we are asked to classify an image with the type of bird. We simply need the information or knowledge about birds.

Information (data)



How do we solve tackle these challenging AI tasks?



Searching Algorithms for AI

Question

- A farmer wants to get his cabbage, goat, and wolf across a river. He has a boat that only holds two. He cannot leave the cabbage and goat alone or the goat and wolf alone. How many river crossings does he need?

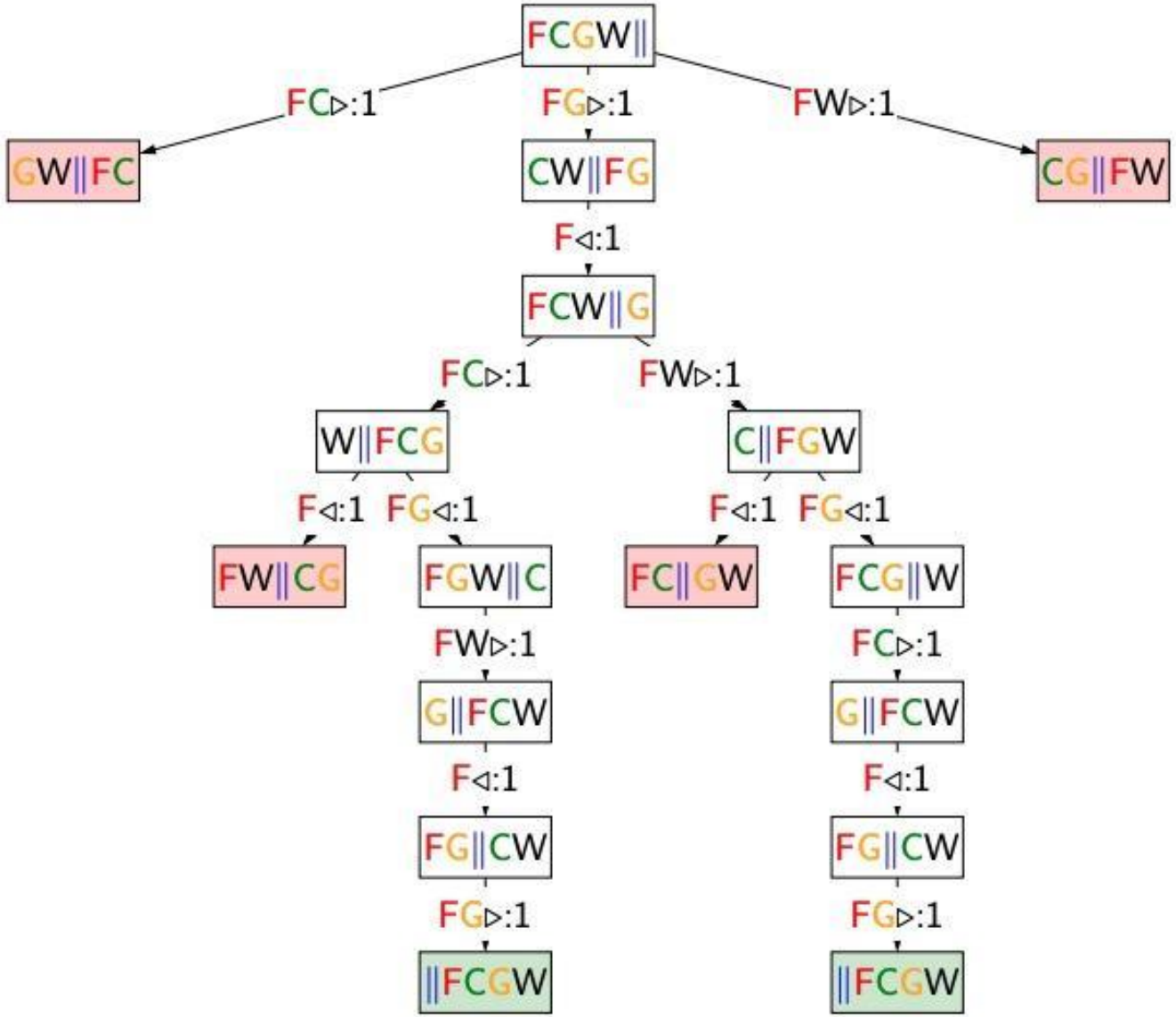
4
5
6
7
no solution



Search Tree

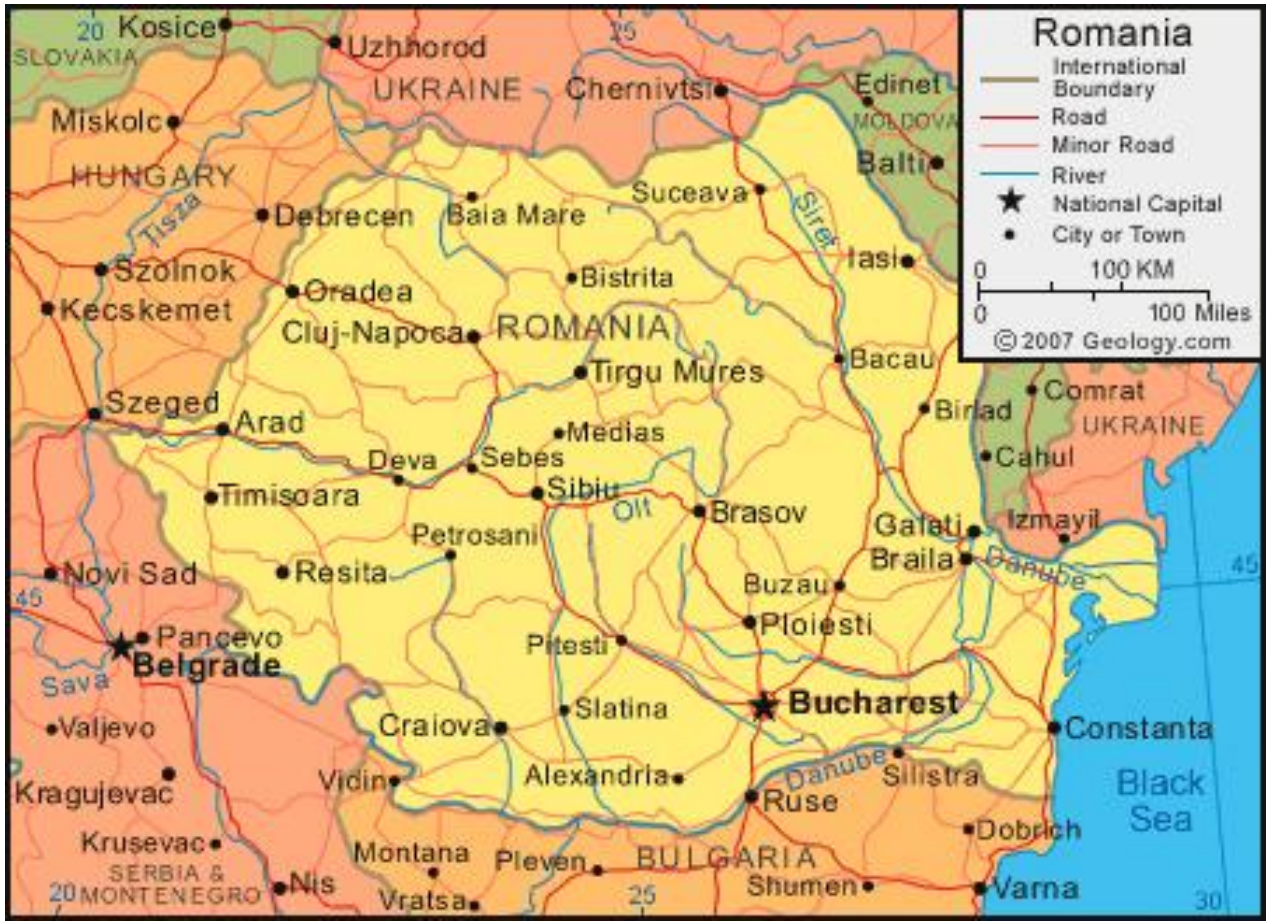


Farmer Cabbage Goat Wolf



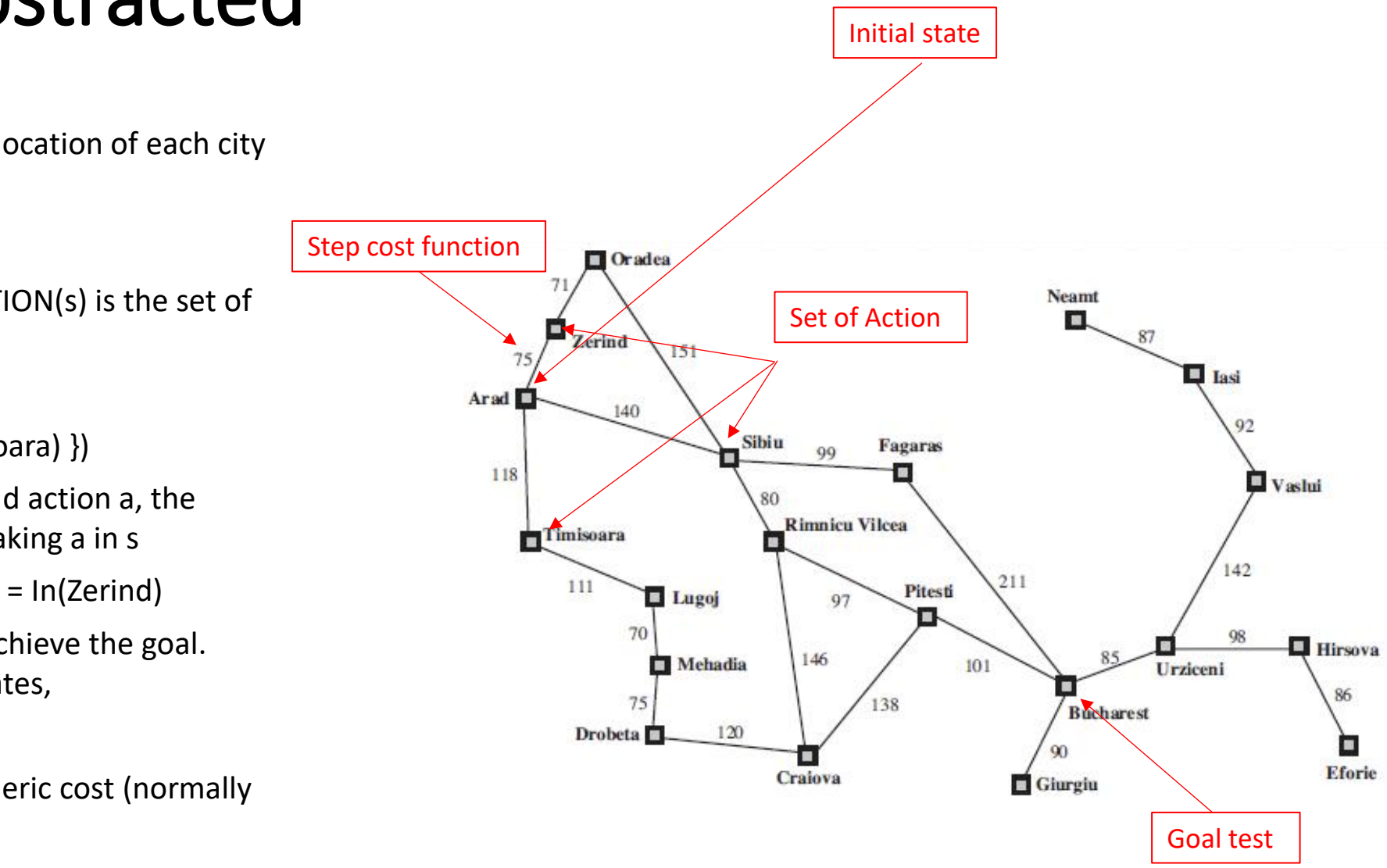
Example: how to get from Arad to Bucharest?

- **Goal formulation**: achieved if location is Bucharest
- **Problem formulation**: various roads of various lengths lead to cities intermediate to Bucharest
- **Search**: find a sequence of intermediate cities with “optimal distance” to Bucharest



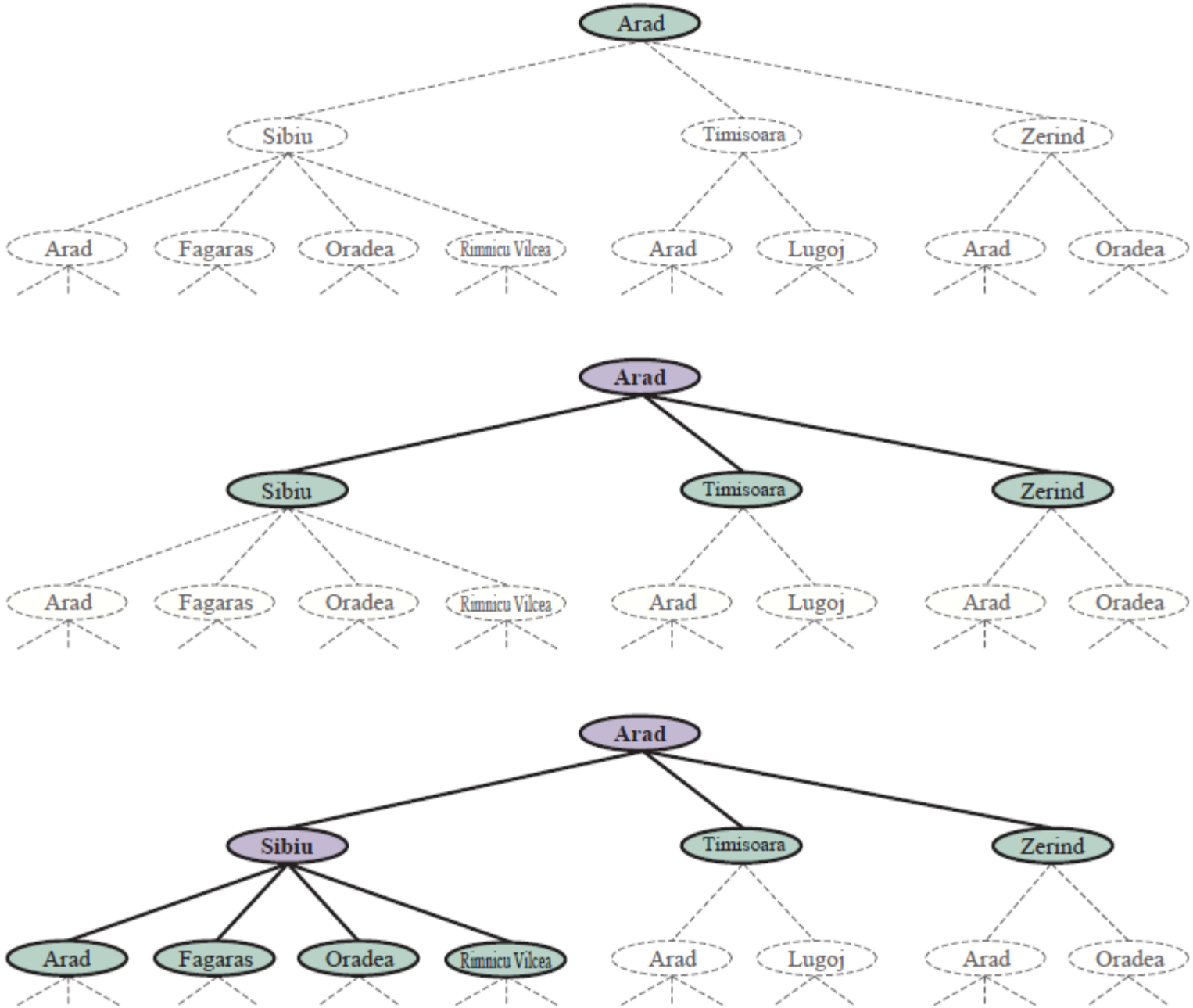
The Problem Abstracted

- **State:** A state description specifies the location of each city
- **An initial state (s)**
  - e.g., In(Arad)
- **A set of actions (a):** given a state s, ACTION(s) is the set of actions applicable in s
  - e.g., ACTION(In(Arad))
  - = {Go(Zerind), Go(Sibiu), Go(Timișoara) })
- **A transition function**: given a state s and action a, the successor RESULT(s, a) is the result of taking a in s
  - e.g., RESULT(In(Arad), Go(Zerind)) = In(Zerind)
- **The goal test** determines what states achieve the goal. Often, there is an explicit set of goal states,
  - e.g., {In(Bucharest)}
- **A step cost function** that assigns a numeric cost (normally non-negative) to each path
  - e.g., c(In(Arad),go(Zerind), In(Zerind)) = 75

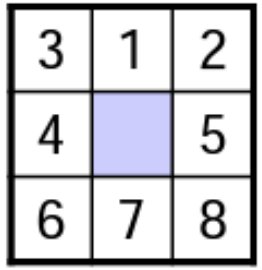


The Problem Abstracted

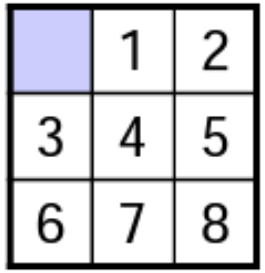
- The **state space** of the problem is the set of possible states of the problem.
  - It is implicitly defined by: the initial state, the action set, and the transition function.
  - The state space forms a directed *graph* in which the nodes are states and the links between nodes are actions.
  - A **path** is a sequence of states connected by a sequence of actions. It has a **path cost** calculated by adding step costs.
  - A **solution** is a path from initial to goal state. It is **optimal** if it has minimal cost.


Arad
Sibiu
Timisoara
Zerind
Arad
Fagaras
Oradea
Rimnicu Vilcea
Arad
Lugoj
Arad
Oradea
Arad
Sibiu
Timisoara
Zerind
Arad
Fagaras
Oradea
Rimnicu Vilcea
Arad
Lugoj
Arad
Oradea
Arad
Sibiu
Timisoara
Zerind
Arad
Fagaras
Oradea
Rimnicu Vilcea
Arad
Lugoj
Arad
Oradea

Example: Slide 8-puzzle



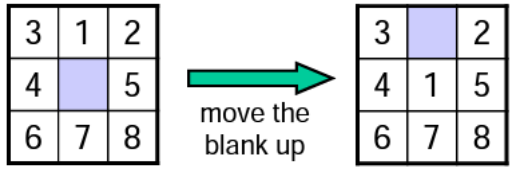
Initial state



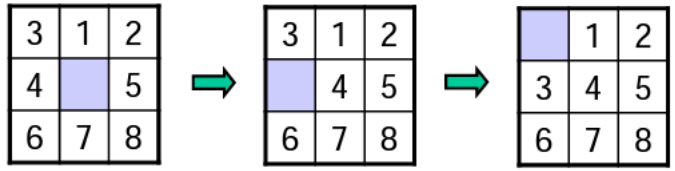
Goal state

- **State**: all configurations of numbers 1-8, plus blank tie.
- **Initial state:** any state can be initial.
- **Actions**: {Left, Right, Up, Down}
- **Goal test**: state matches goal configuration
- **Step cost**: each step costs 1 (i.e., path cost =#actions in the path)
- **Transition function:**

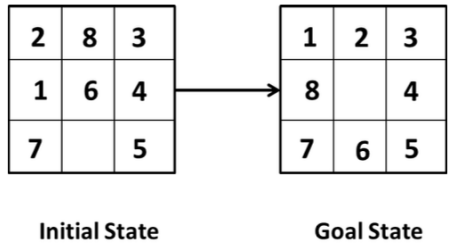
e.g.,

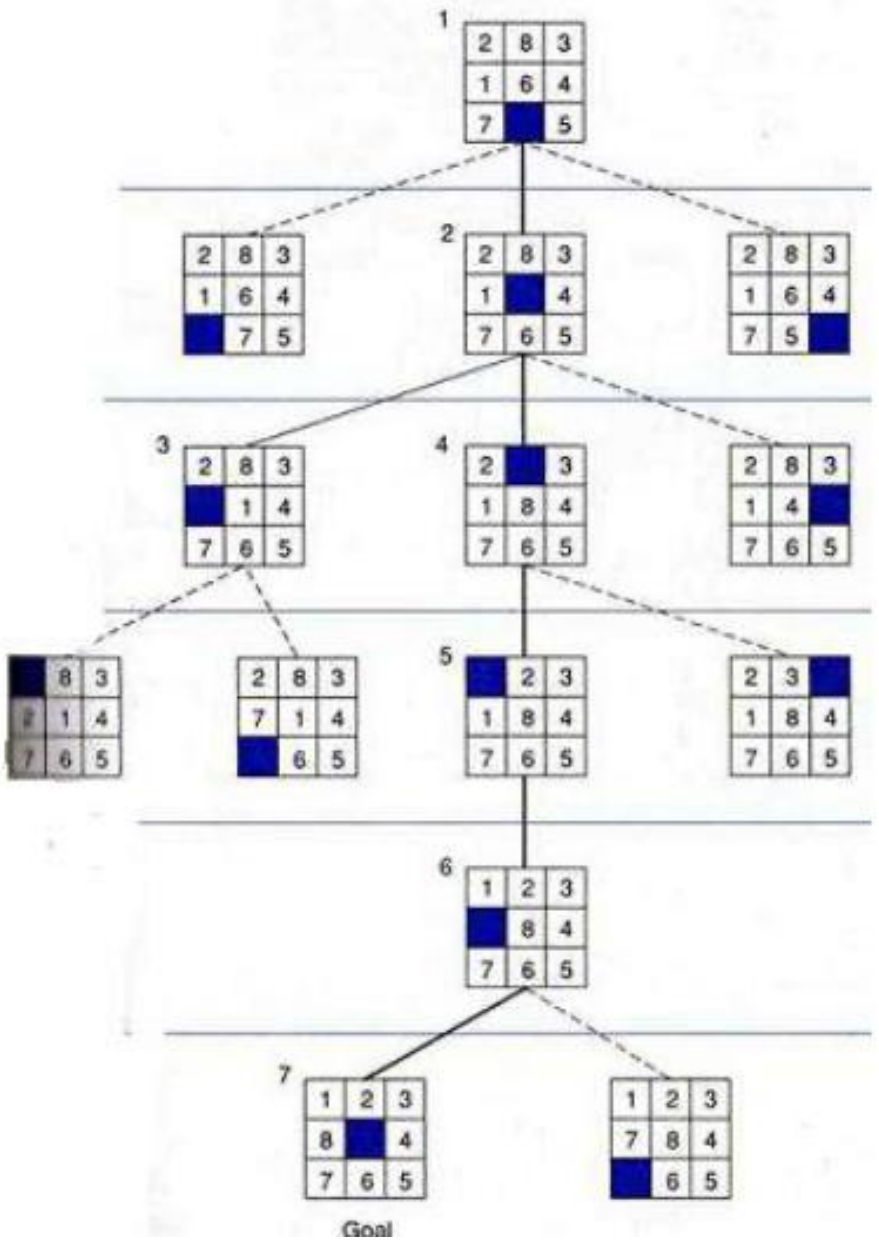


How to search for solutions?



Example: Slide 8-puzzle





How to search for solutions?



- Two Search Types:
  - **Uninformed search (blind search)**
    - ✓ No information about the number of steps or the path cost from the current state to the goal.
    - ✓ All they can do is distinguish a goal state from a non-goal state.
  - **Informed search (heuristic search)**
    - ✓ Use some information about a goal state.

Search Strategies Performance Measure

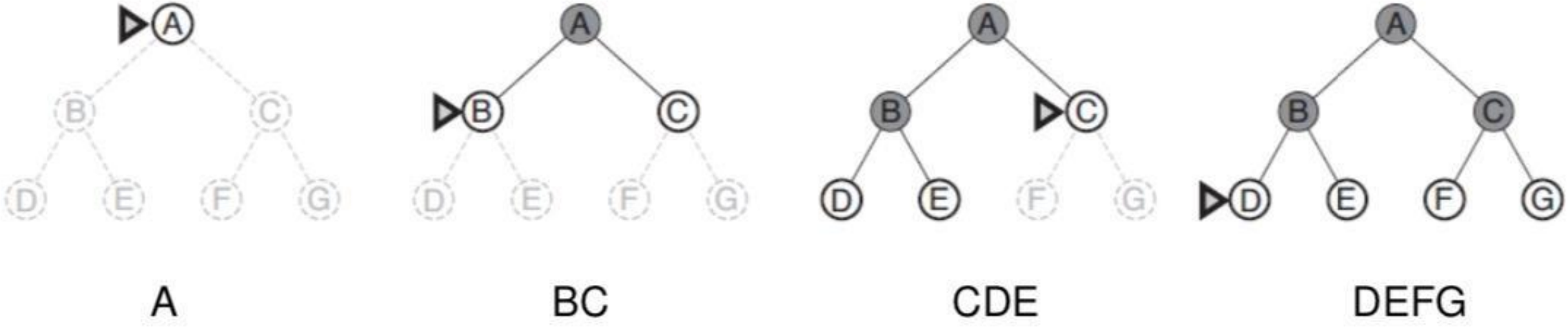
- A search strategy is defined by picking the order of node expansion.
- Strategies are evaluated along the following dimensions:
  - Completeness: does it always find a solution if one exists?
  - Time complexity: number of nodes generated.
  - Space complexity: maximum number of nodes in memory.
  - Optimality: does it always find a least-cost solution?
  - Systematicity: does it visit each state at most once?
- Time and space complexity are measured in terms of
  - b: maximum branching factor of the search tree.
  - d: depth of the least-cost solution.
  - m: maximum depth of the state space (maybe ∞)



Uninformed Search

1. Breadth-first Search (BFS)

- All the nodes at depth d in the search tree are expanded before the nodes at depth d+1.



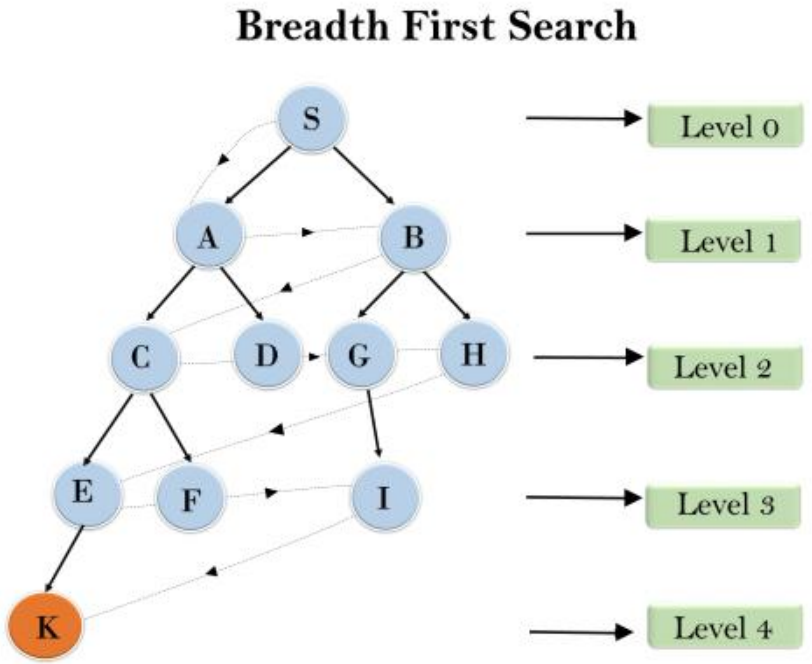
1. Breadth-first Search (BFS)

Advantage

- If there is a solution, breadth-first search is guaranteed to find it.
- If there are several solutions, breadth-first search will always find the shallowest goal state first.

Drawbacks

- All the leaf nodes of the tree must be maintained in memory at the same time, resulting in enormous memory consuming and time consuming in search.



1. Breadth-first Search (BFS)

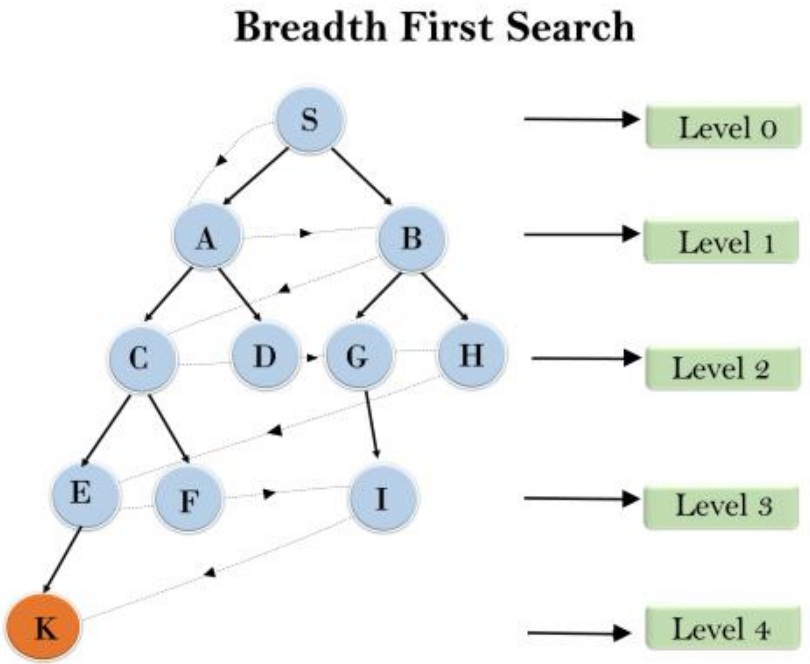
Time vs Memory

Depth	Nodes	Time	Memory
2	110	.11 milliseconds	107 kilobytes
4	11,110	11 milliseconds	10.6 megabytes
6	10^6	1.1 seconds	1 gigabyte
8	10^8	2 minutes	103 gigabytes
10	10^{10}	3 hours	10 terabytes
12	10^{12}	13 days	1 petabyte
14	10^{14}	3.5 years	99 petabytes
16	10^{16}	350 years	10 exabytes

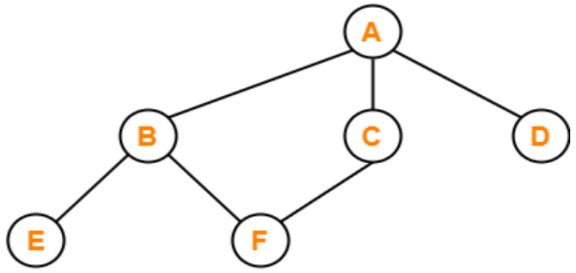
Figure 3.13 Time and memory requirements for breadth-first search. The numbers shown assume branching factor $b = 10$; 1 million nodes/second; 1000 bytes/node.

1. Breadth-first Search (BFS)

- Completeness: Yes
- Time complexity $O(b^d)$
- Space complexity $O(b^d)$
- Cost optimality: Not Guaranteed



Example



In [1]:

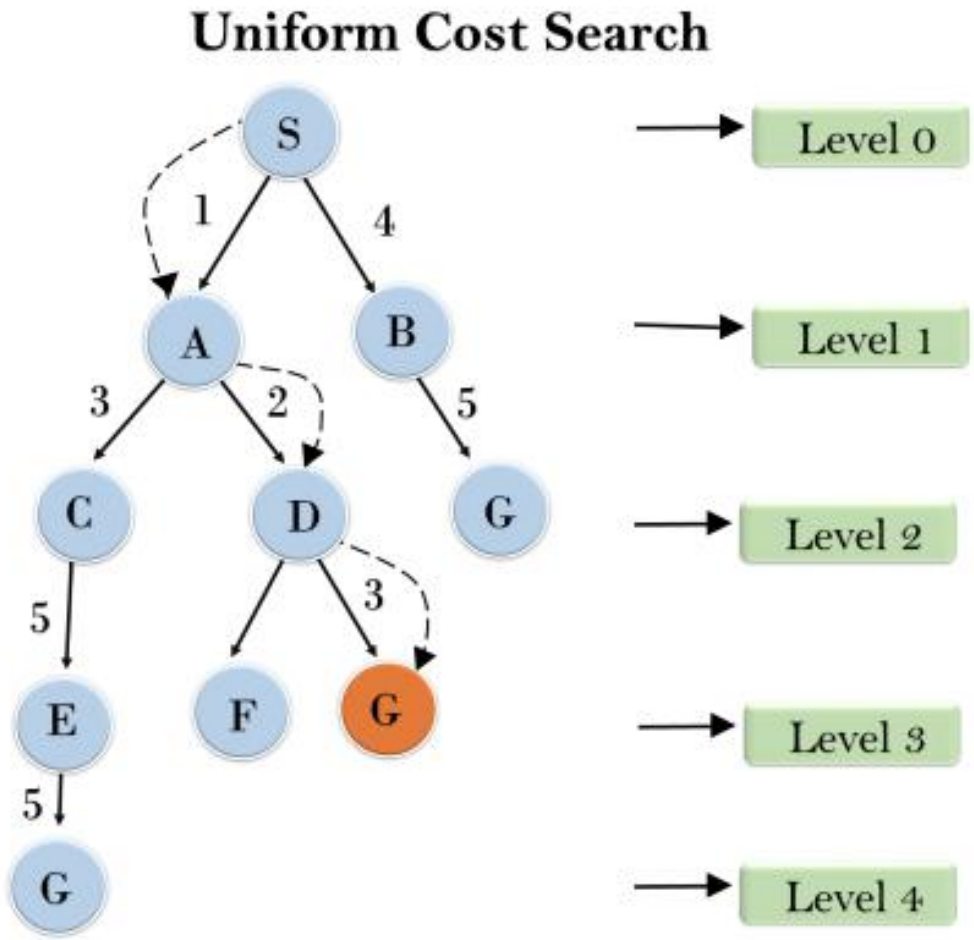
```
from collections import defaultdict
class Graph:
    def __init__(self):
        self.graph=defaultdict(list)
    def addEdge(self, u, v):
        self.graph[u].append(v)
    def BFS(self, s):
        visited = [False]*7
        queue=[]
        queue.append(s)
        visited[s]=True
        while queue:
            s=queue.pop(0)
            print(s, end="")
            for i in self.graph[s]:
                if visited[i]==False:
                    queue.append(i)
                    visited[i]=True

g = Graph()
g.addEdge(1, 2)
g.addEdge(1, 3)
g.addEdge(1, 4)
g.addEdge(2, 5)
g.addEdge(2, 6)
g.addEdge(3, 6)
print ("Follwing is Breath First Traversal (Starting from vertex 1)")
g.BFS(1)
# ['A':1, 'B':2, 'C':3, 'D':4, 'E':5, 'F':6]
```

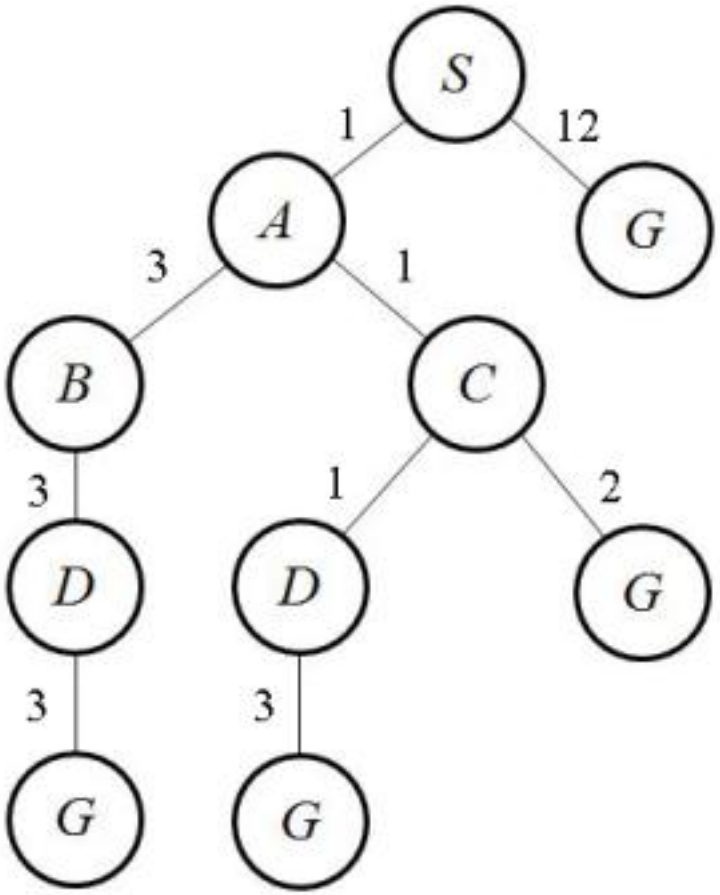
```
Follwing is Breath First Traversal (Starting from vertex 1)
123456
```

2. Uniform-cost search

- Breadth-first search finds the shallowest goal state, but this may not always be the least-cost solution for a general path cost function.
- Uniform cost search modifies the breadth-first strategy by always **expanding the lowest-cost node**.
- It does not care about the number of steps involved in searching and only concerned about path cost.



Example

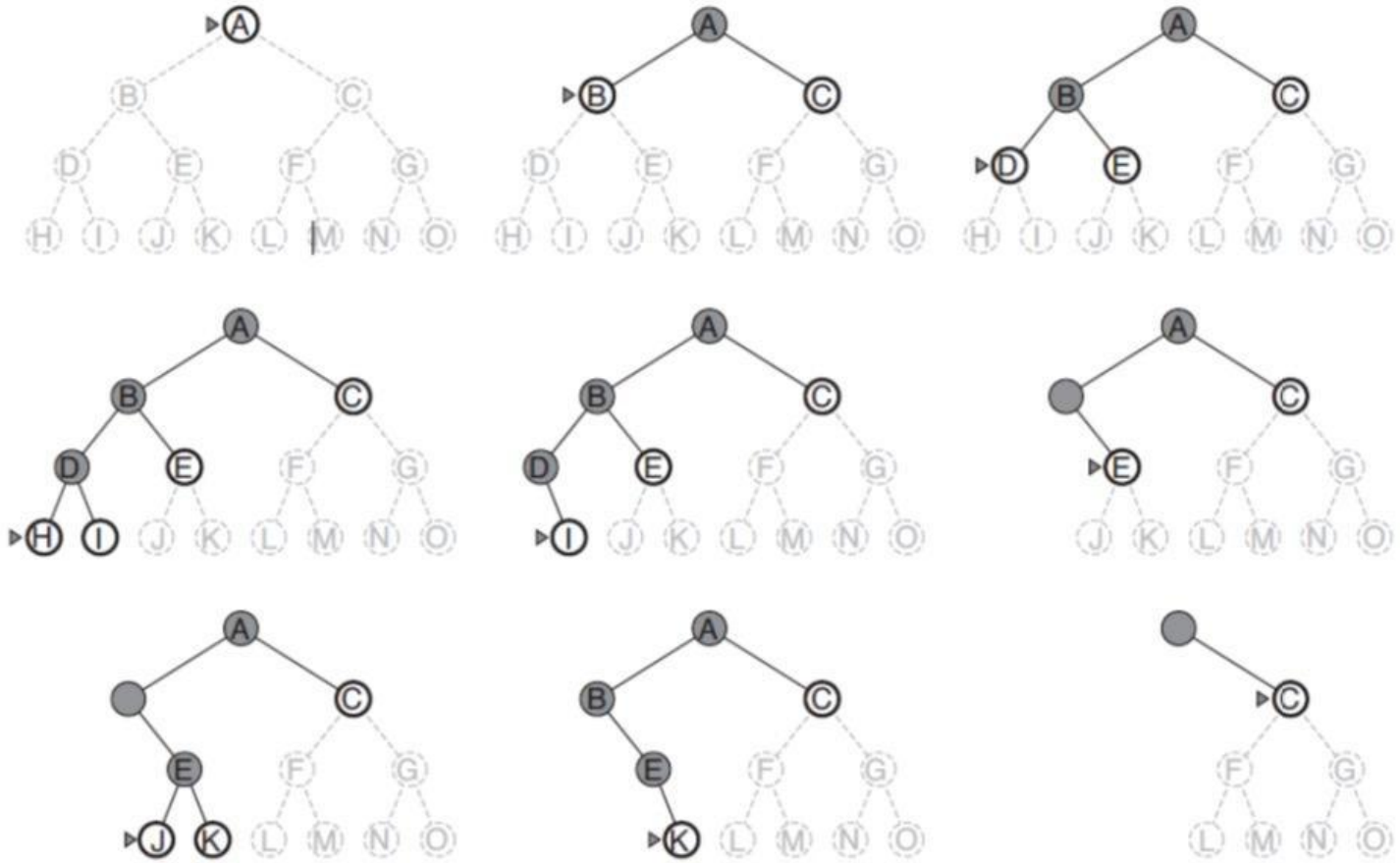


Insert the root into the queue
While the queue is not empty
Dequeue the maximum priority element from the queue
(If priorities are same, alphabetically smaller path is chosen)
If
the path is ending in the goal state, print the path and exit
Else
Insert all the children of the dequeued element, with the cumulative costs as priority

Initialization: { [S , 0] }
Iteration1: { [S->A , 1] , [S->G , 12] }
Iteration2: { [S->A->C , 2] , [S->A->B , 4] , [S->G , 12] }
Iteration3: { [S->A->C->D , 3] , [S->A->B , 4] , [S->A->C->G , 4] , [S->G , 12] }
Iteration4: { [S->A->B , 4] , [S->A->C->G , 4] , [S->A->C->D->G , 6] , [S->G , 12] }
Iteration5: { [S->A->C->G , 4] , [S->A->C->D->G , 6] , [S->A->B->D , 7] , [S->G , 12] }
Iteration6 gives the final output as S->A->C->G.

3. Depth-first search (DFS)

- Depth-first search always expands one of the nodes at the deepest level of the tree.



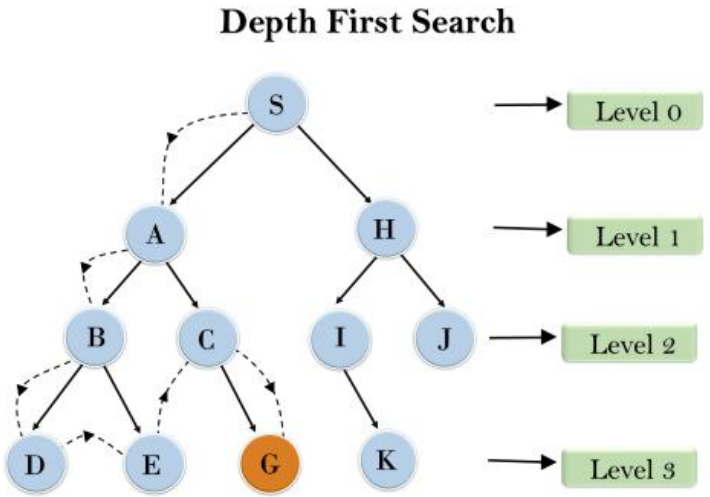
3. Depth-first search (DFS)

Advantage

- Depth-first search has very modest memory requirements
  - It needs to store only a single path from the root to a leaf node, along with the remaining unexpanded sibling nodes for each node on the path.

Drawbacks

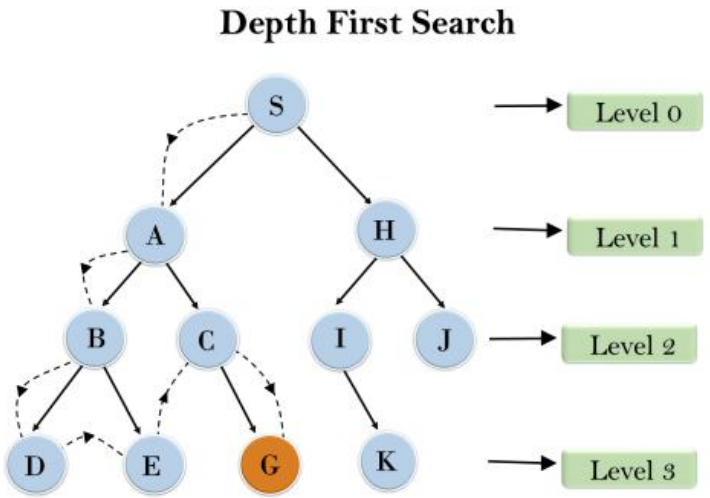
- It can get stuck going down the wrong path
  - Many problems have very deep or even infinite search trees;
  - The search will always continue downward without backing up, even when a shallow solution exists.



Root node--->Left node ----> right node.

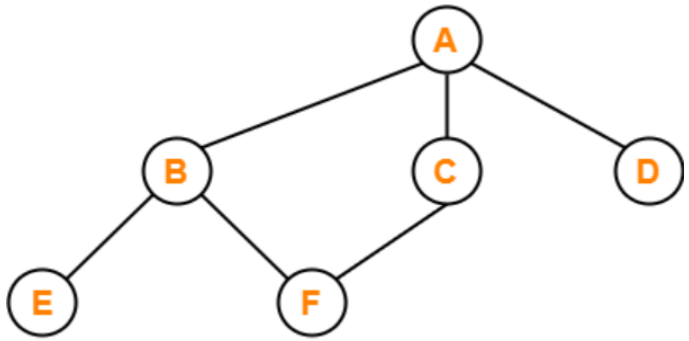
3. Depth-first search (DFS)

- Completeness: Not Guaranteed
- Time complexity $O(b^m)$
- Space complexity $O(bm)$
- Cost optimality: Not Guaranteed



Root node--->Left node ----> right node.

Example:



In [6]:

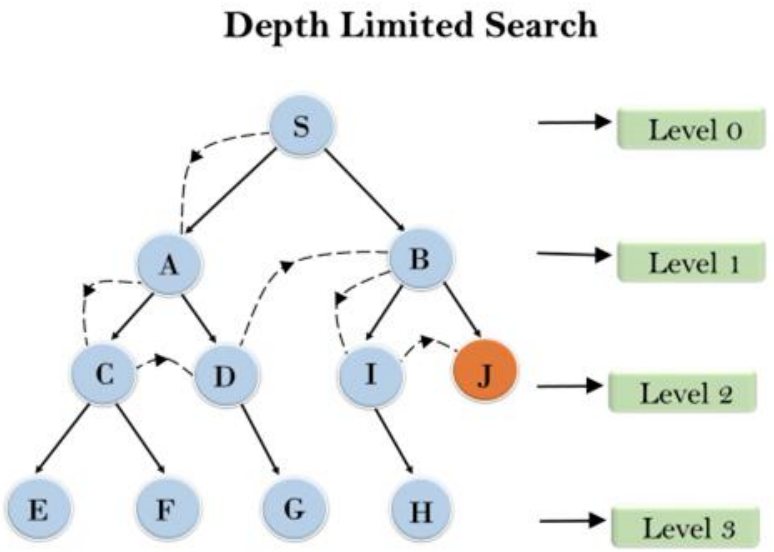
```
from collections import defaultdict
class Graph:
    def __init__(self):
        self.graph = defaultdict(list)
    def addEdge(self, u, v):
        self.graph[u].append(v)
    def DFSUtil(self, v, visited):
        visited.add(v)
        print(v, end=' ')
        for neighbour in self.graph[v]:
            if neighbour not in visited:
                self.DFSUtil(neighbour, visited)
    def DFS(self, v):
        visited = set()
        self.DFSUtil(v, visited)
g = Graph()
g.addEdge(1, 2)
g.addEdge(1, 3)
g.addEdge(1, 4)
g.addEdge(2, 5)
g.addEdge(2, 6)
g.addEdge(3, 6)

print("Following is DFS from (starting from vertex 1)")
g.DFS(1)
# {'A':1, 'B':2, 'C':3, 'D':4, 'E':5, 'F':6}
```

```
Following is DFS from (starting from vertex 1)
1 2 5 6 3 4
```

4. Depth-limited search

- Depth-limited search is an extension of DFS:
  - only expands nodes with depth up to *l* .
- For example, on the map of Romania, there are 20 cities, so we know that if there is a solution, then it must be of length 19 at the longest.
- Hard to choose an appropriate depth number *l*. For most problems, we will not know a good depth limit until we have solved the problem.



4. Depth-limited search

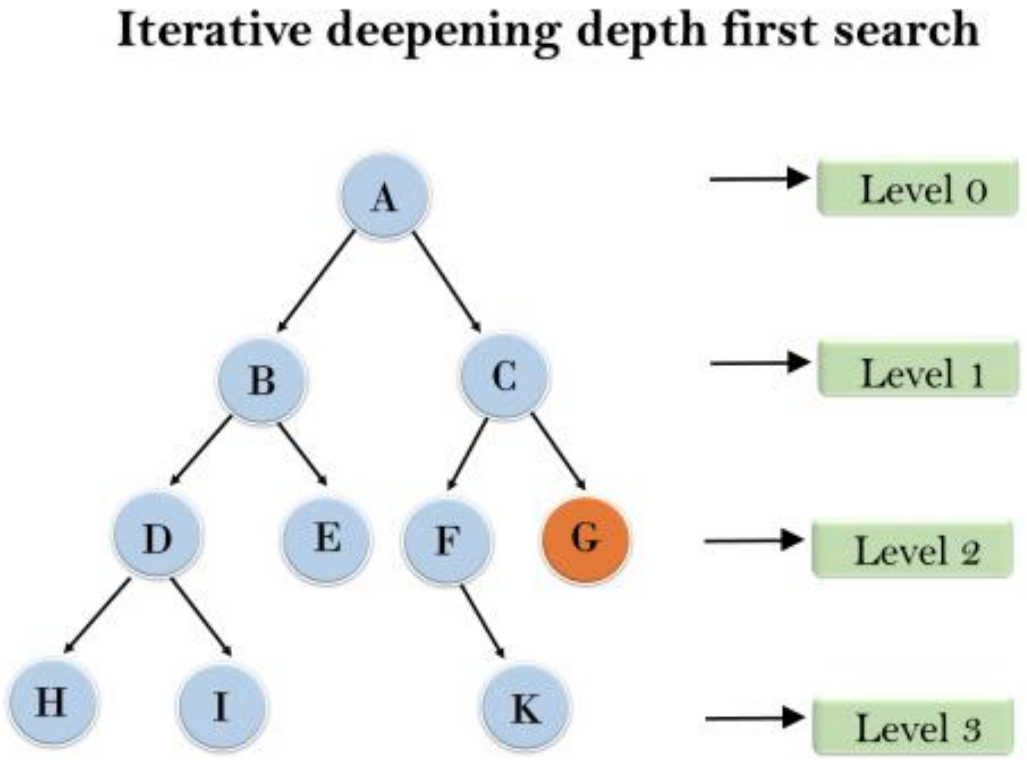
- Advantages:
  - It is memory efficient
  - Space complexity $O(bl)$
  - Time complexity $O(b^l)$
- Disadvantages
  - Incompleteness.
  - It may not be optimal if the problem has more than one solution.

5. Iterative deepening search

- The iterative deepening algorithm is a combination of DFS and BFS algorithms. This search algorithm finds out the best depth limit and does it by **gradually increasing** the limit until a goal is found.

- This algorithm performs depth-first search up to a certain "depth limit", and it keeps increasing the depth limit after each iteration until the goal node is found.

- The main drawback is that it repeats all the work of the previous phase.



1'st Iteration-----> A

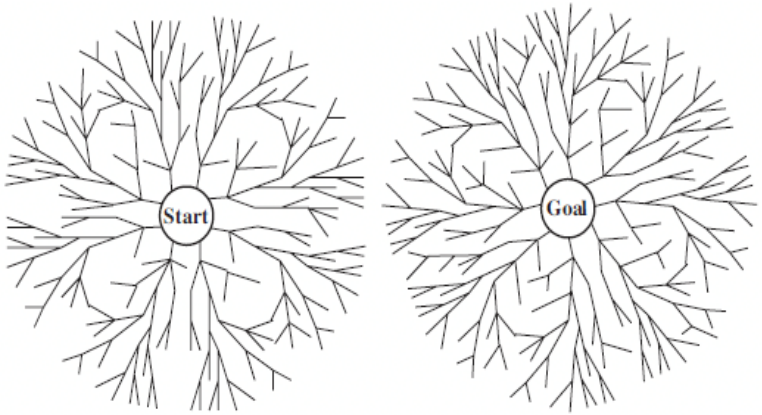
2'nd Iteration----> A, B, C

3'rd Iteration------>A, B, D, E, C, F, G

4'th Iteration------>A, B, D, H, I, E, C, F, K, G

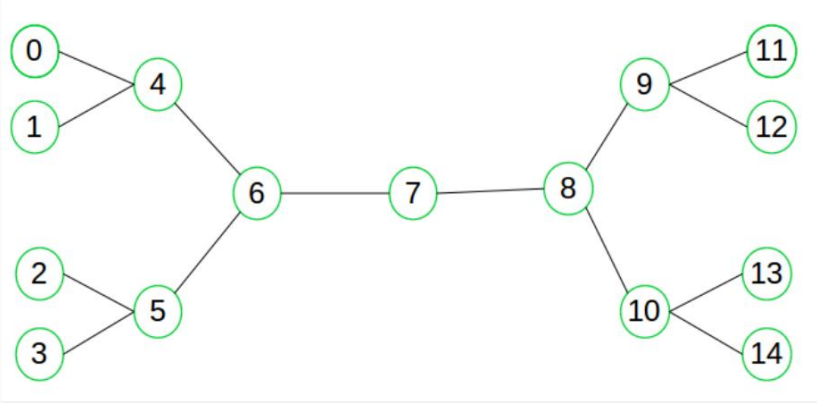
6. Bidirectional search

- Bidirectional search algorithm runs **two simultaneous searches**, one form initial state called as forward-search and other from goal node called as backward-search, to find the goal node.



- Bidirectional search replaces one single search graph with two small subgraphs in which one starts the search from an initial vertex and the other starts from the goal vertex.

- The search stops when these two graphs intersect each other.



- Bidirectional search can use search techniques such as BFS, DFS, DLS, etc.

- In bidirectional search, one should know the goal state in advance.

Comparisons

Criterion	Breadth-First	Uniform-Cost	Depth-First	Depth-Limited	Iterative Deepening	Bidirectional (if applicable)
Complete?	Yes[1]	Yes[1,2]	No	No	Yes[1]	Yes[1,4]
Optimal cost?	Yes[3]	Yes	No	No	Yes[3]	Yes[3,4]
Time	$O(b^d)$	$O(b^{1+\lfloor C^*/\epsilon \rfloor})$	$O(b^m)$	$O(b^\ell)$	$O(b^d)$	$O(b^{d/2})$
Space	$O(b^d)$	$O(b^{1+\lfloor C^*/\epsilon \rfloor})$	$O(bm)$	$O(b\ell)$	$O(bd)$	$O(b^{d/2})$

Figure 3.15 Evaluation of search algorithms. b is the branching factor; m is the maximum depth of the search tree; d is the depth of the shallowest solution, or is m when there is no solution; ℓ is the depth limit. Superscript caveats are as follows: [1] complete if b is finite, and the state space either has a solution or is finite. [2] complete if all action costs are $\geq \epsilon > 0$; [3] cost-optimal if action costs are all identical; [4] if both directions are breadth-first or uniform-cost.



Informed Search

Best-first Search

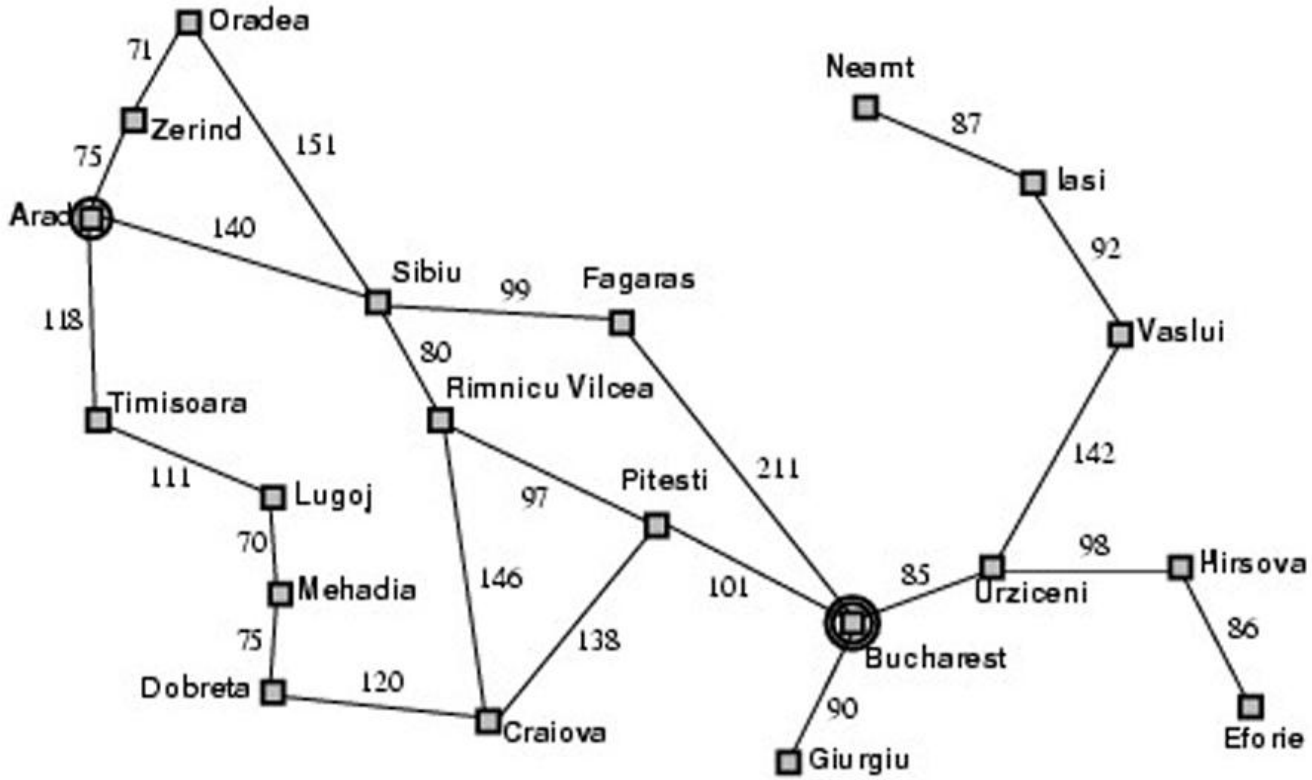
- Provide an evaluation function that returns a number purporting to describe the desirability
- Best-first Search
  - Use an **evaluation functio**n for each node.
  - Estimate of "desirability".
  - Expand most desirable unexpanded node.
- There is a whole family of Best-first Search algorithms with different evaluation functions
  - Two basic approaches
    - ✓ The first (named *Greedy search*) tries to expand the node closest to the goal;
    - ✓ The second (named *A* search*) tries to expand the node on the least-cost solution path.

Greedy Search: Minimize Estimated Cost to Reach a Goal

- For most problems, the cost of reaching the goal from a particular state can be estimated but cannot be determined exactly.
- A function that calculates such cost estimates is called a heuristic function (usually denoted by the letter *h*):
  - h(n) = estimated cost of **the cheapest path from the state at node n to a goal state**
  - A best-first search that uses h to select the next node to expand is called *greedy search*.
  - Strictly speaking, *h* can be any function at all. We will require only that *h(n) = 0* if n is a goal.

An Example - Route-finding Problems

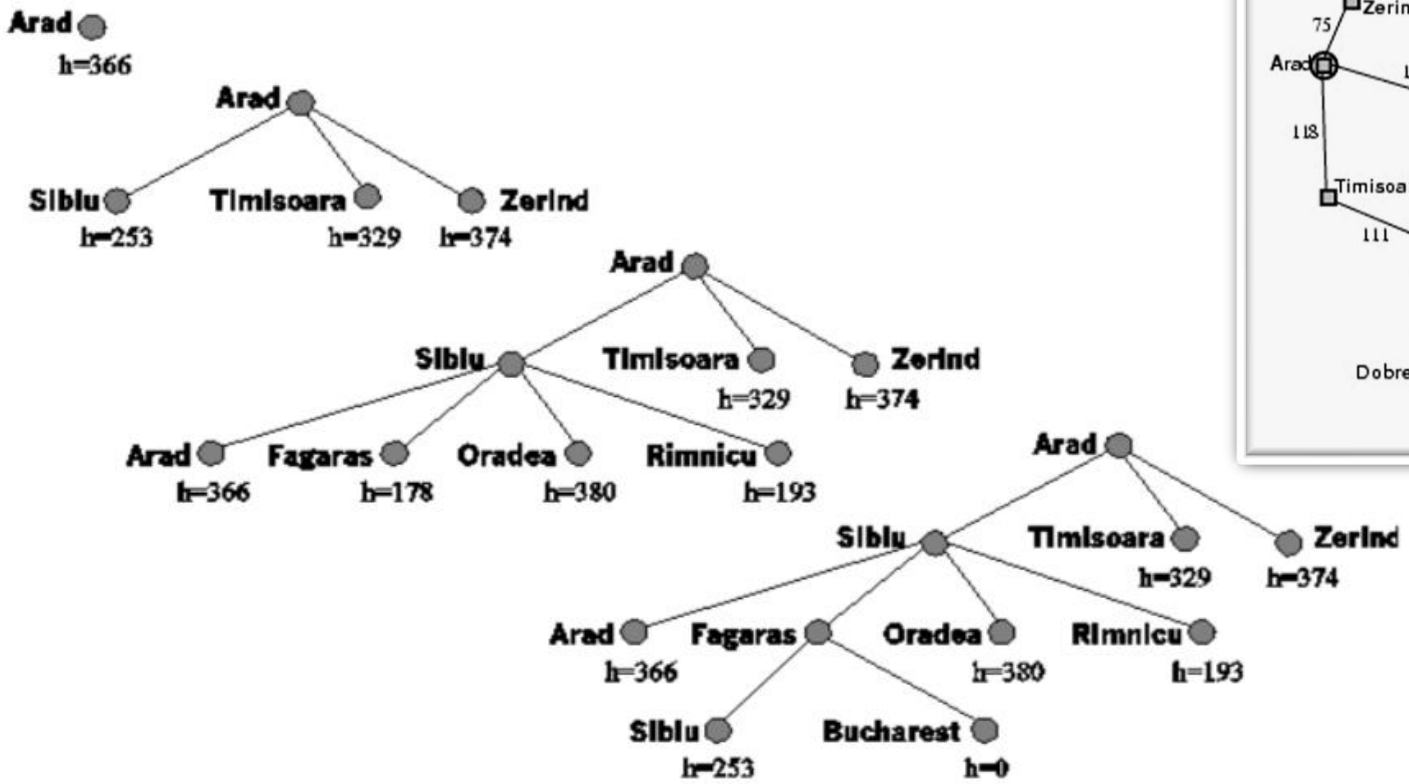
What's a proper heuristic that measures cheapest path from current node to goal node?

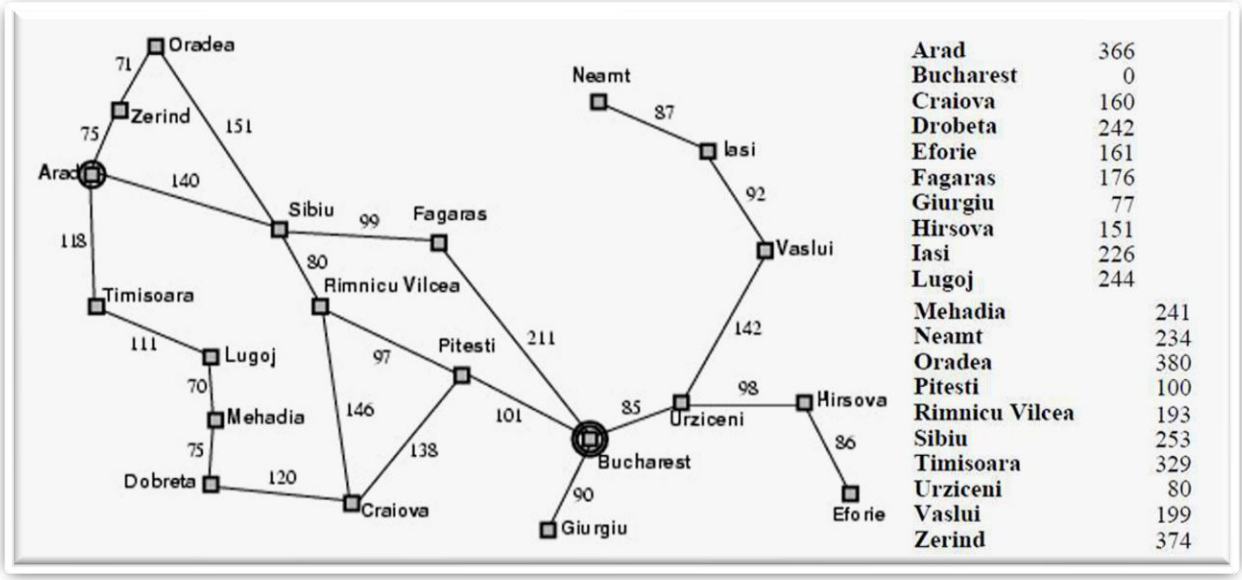


Arad	366
Bucharest	0
Craiova	160
Drobeta	242
Eforie	161
Fagaras	176
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	100
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

$h_{SLD}(n)$ = straight line distance (SLD) between n and the goal location

Greedy Search in the Example (From Arad to Bucharest)





Greedy search prefers to take the biggest bite possible out of the remaining cost to reach the goal, without worrying about whether this will be best in the long run.

Result:

Arad ☐ Sibiu ☐ Făgăraș ☐ Bucharest

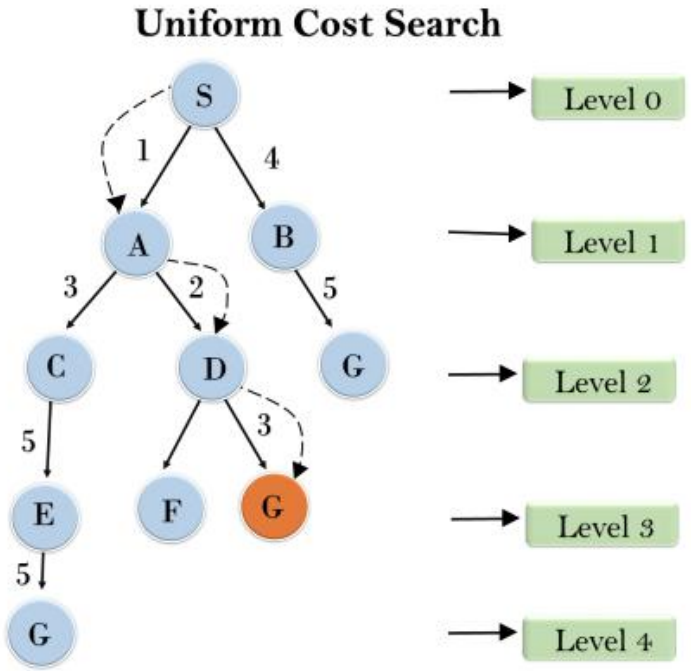
Note:

- The result is not perfectly optimal
- The optimal result should be:

Arad ☐ Sibiu ☐ Râmnicu Vâlcea ☐ Pitesti ☐ Bucharest

Greedy Search V.S. Uniform-cost Search

- Greedy Search
  - Minimize the estimated cost to the goal h(n)
- Uniform-cost Search
  - Minimize the cost the path so far, g(n)



- Can we combine these two strategies to get the advantages of both?

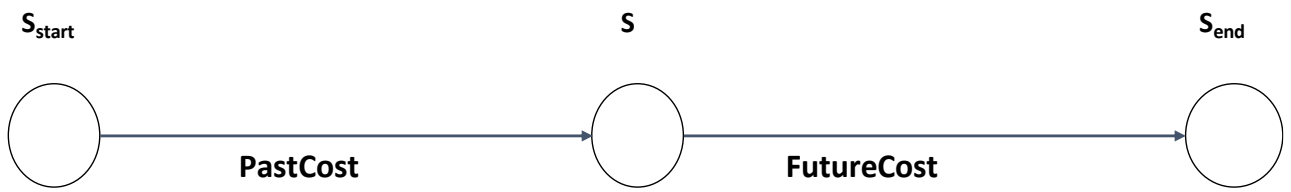
Minimizing the Total Path Cost: **A* Search**

- We can do exactly that, combining the two evaluation functions simply by

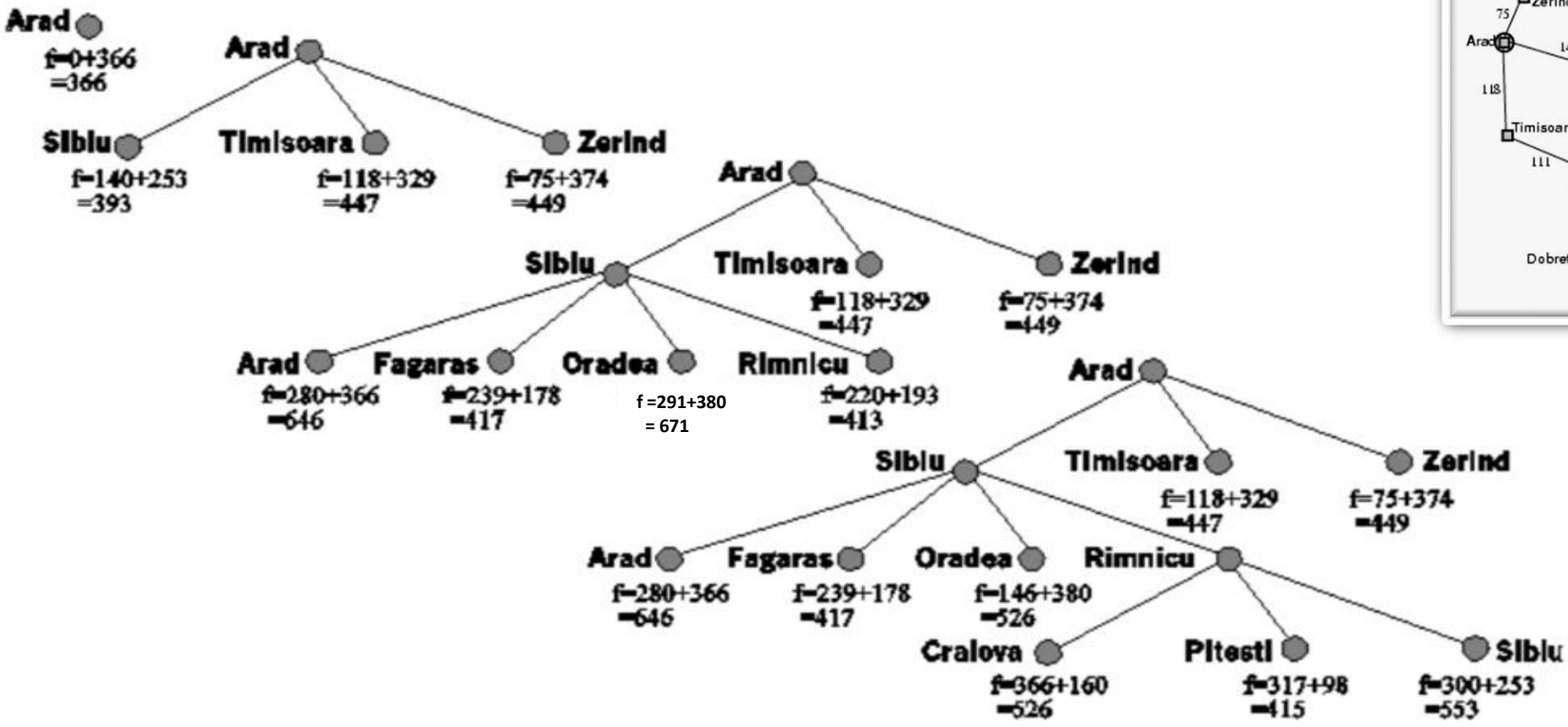
$$f(n) = g(n) + h(n)$$

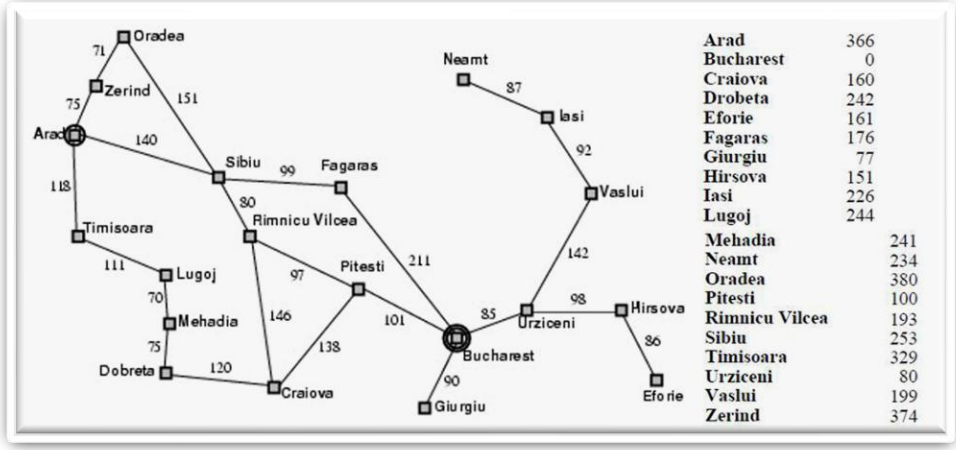
where:

- g(n): the path cost from the start node to node n;
- h(n): the estimated cost of the cheapest path from n to the goal;
- We have: f(n) = estimated cost of the cheapest solution through n.



A* Search in a Route-finding Problem



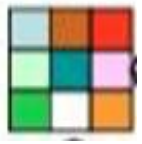


The Behavior of A* Search

- A heuristic for which it holds is said to exhibit **monotonicity**.
  - Let us consider two nodes n and n', where n is the parent of n'.
  - Now suppose, for example, that g(n) = 3 and h(n) = 4, then f(n) = g(n) + h(n) = 7.
    - ✓ It means that the true cost of a solution path through n is at least 7.
  - Suppose also that g(n') = 4 and h(n') = 2, then f(n') = 6.
  - (an example of a nonmonotonic heuristic)
- If the heuristic is one of those odd ones that is not monotonic, we can make a minor correction that restores monotonicity

$$f(n') = \max(f(n), g(n') + h(n'))$$

How to Find a Good Heuristic Function?



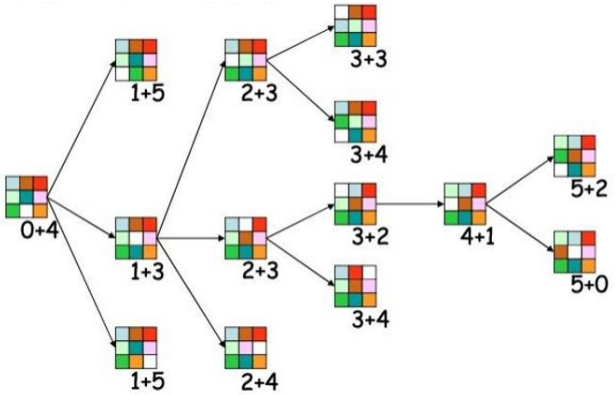
Start state



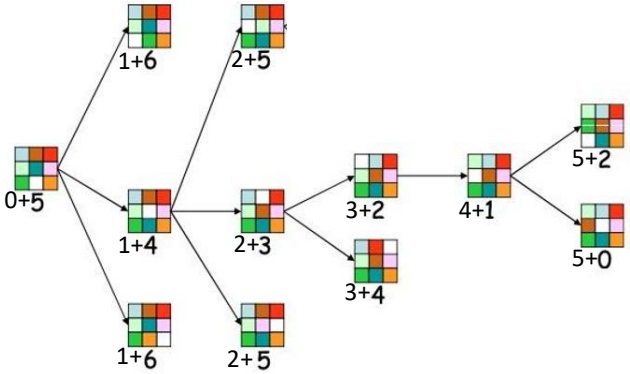
Goal state

h1 = 4
h2 = 5

- Two candidates in this 8 puzzle problem
  - h1 = the number of tiles that are in the wrong positions.
  - h2 = the sum of the distances of the tiles from their goal positions. (also called **city block distance** or **Manhattan distance**)



h1



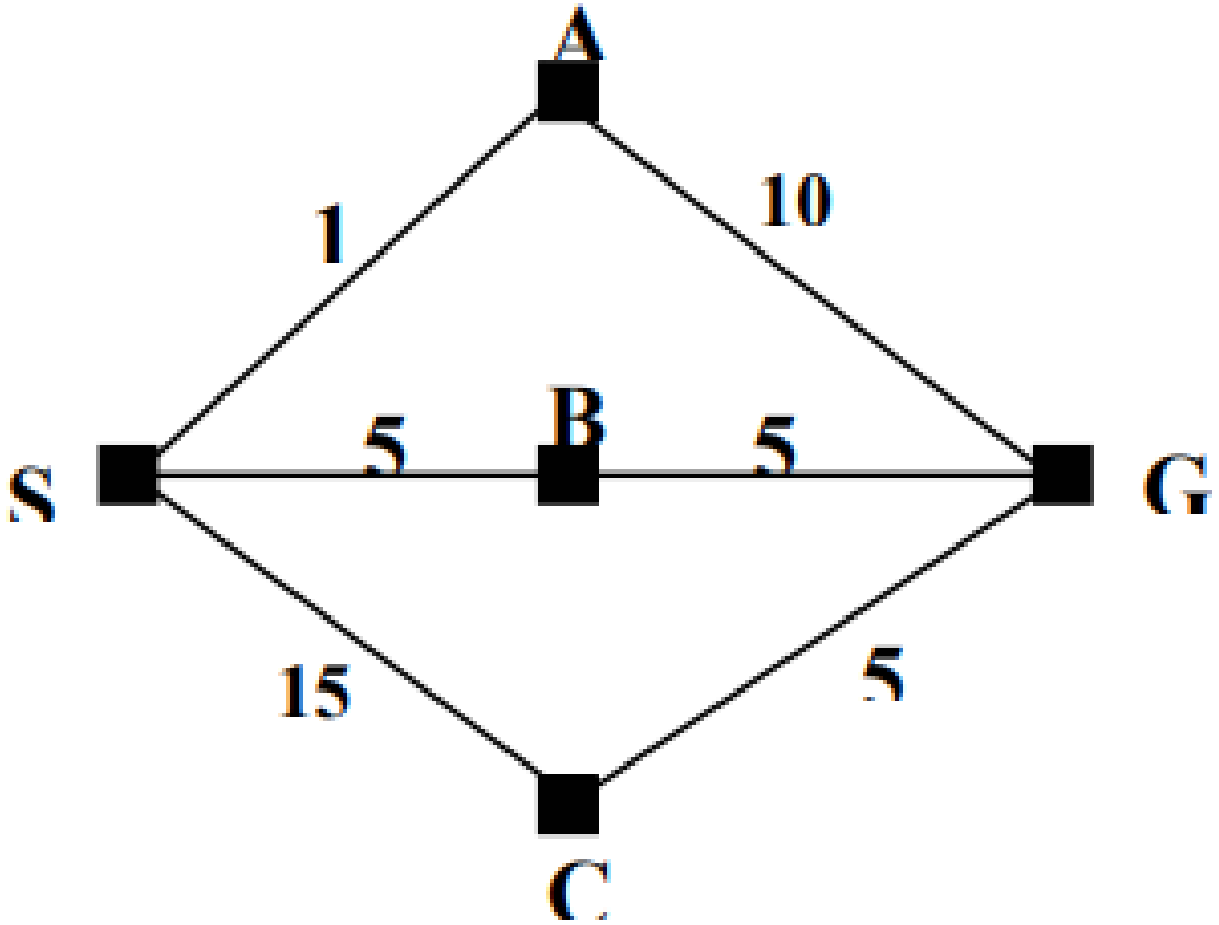
h2

Note that by definition, $h2(n) \geq h1(n)$: $h2$ **dominates** $h1$

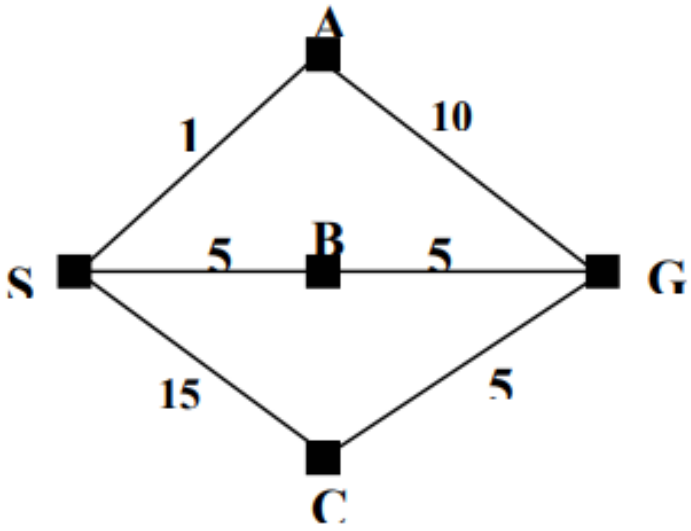
It is always better to use a heuristic function with higher values, as long as it does not overestimate.

Class Question

Question: Show Uniform Cost search Operates



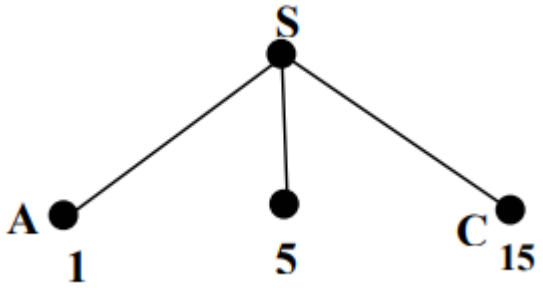
Class Question



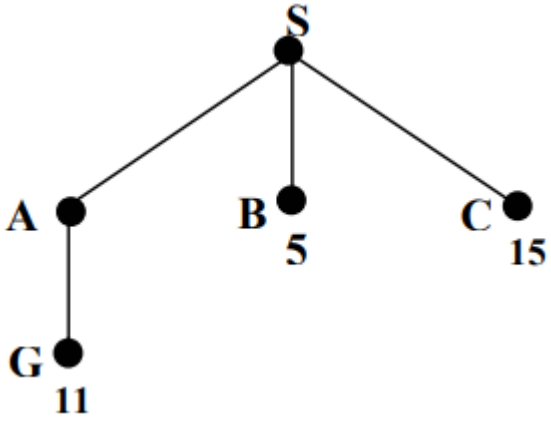
Question: Show Uniform Cost search Operates

Answer:

Step-1: We start with the initial state and expand it

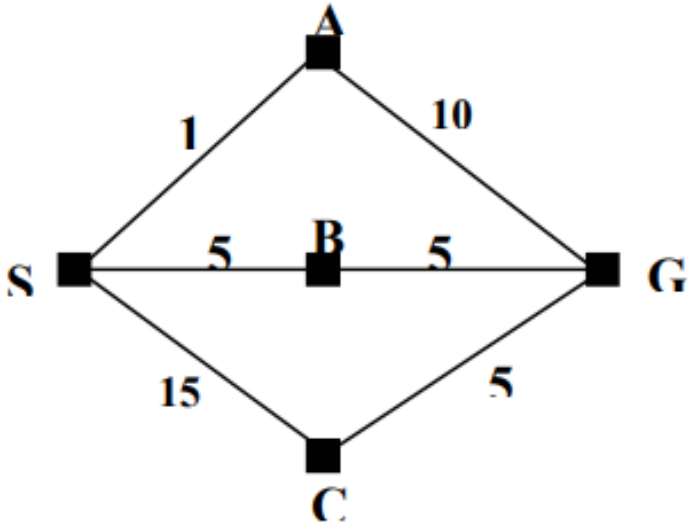


Step-2: The path cost of A is the cheapest, so it is expanded next; giving the following tree



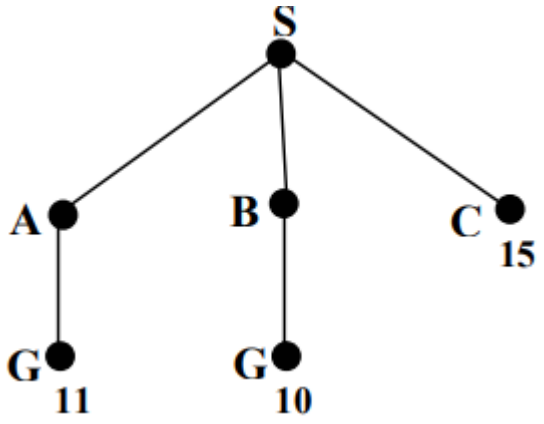
Class Question

Question: Show Uniform Cost search Operates



Step-3: We now have a goal state, but the algorithm does not recognize it yet as there is still a node with a cheaper path cost. In fact, what the algorithm does is order the queue by the path cost so the node with cost 11 will be behind node B in the queue. Node B (being the cheapest) is now expanded, giving

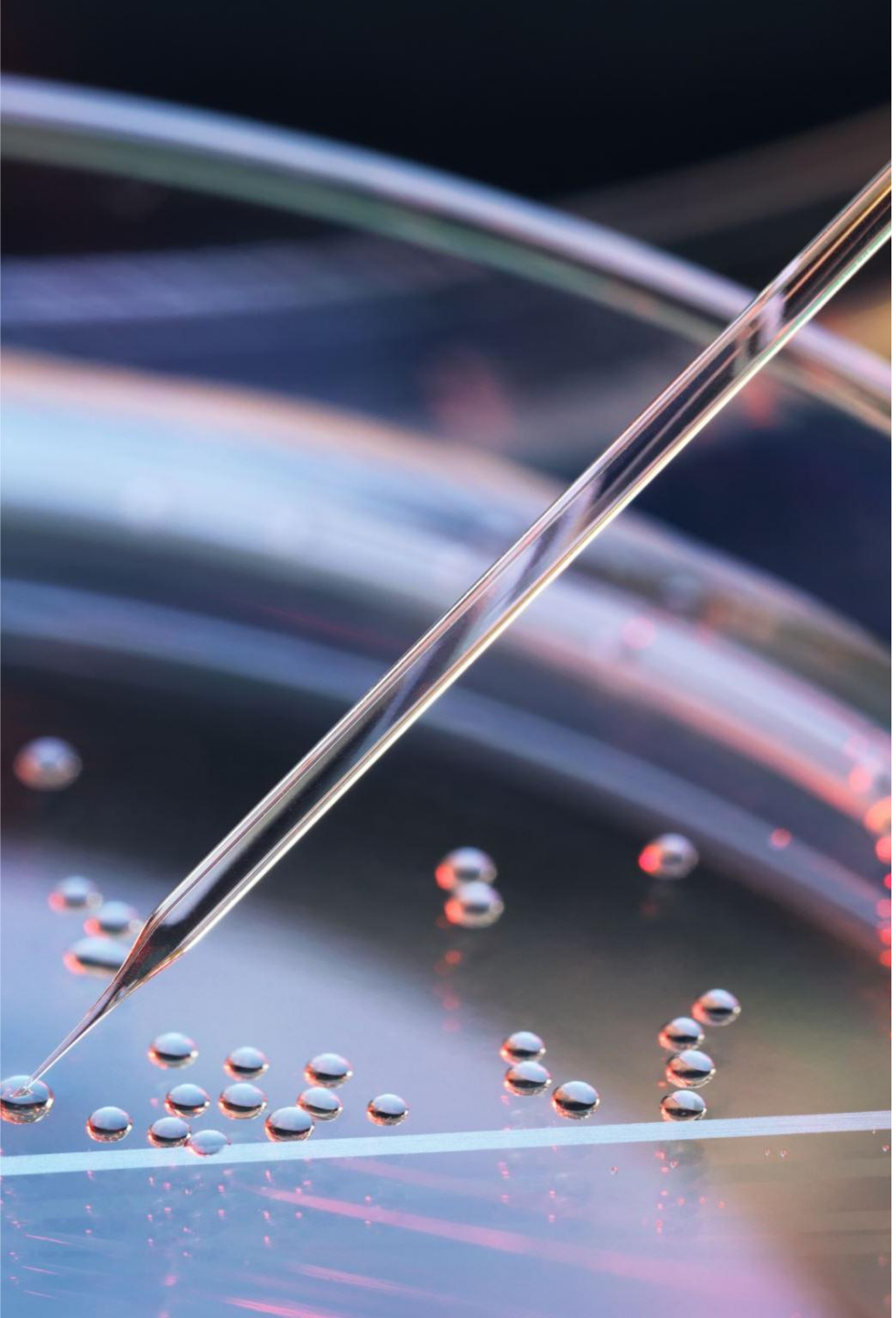
A goal state (G) will now be at the front of the queue. Therefore, the search will end and the path SBG will be returned.



$S \rightarrow B \rightarrow G = 10$

$S \rightarrow A \rightarrow G = 11$

cheapest solution



Summary

- Uninformed search
  - Breath-first search
  - Uniformed-cost search
  - Depth-first search
  - Depth-limited search
  - Iterative deepening search
  - Bidirectional search
  - Backtracking with forwarding checking
- Informed Search
  - Best-first search
    - Greedy search
    - A* search

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  - https://leetcode.com/problems/number-of-islands/